

BFS Traversal

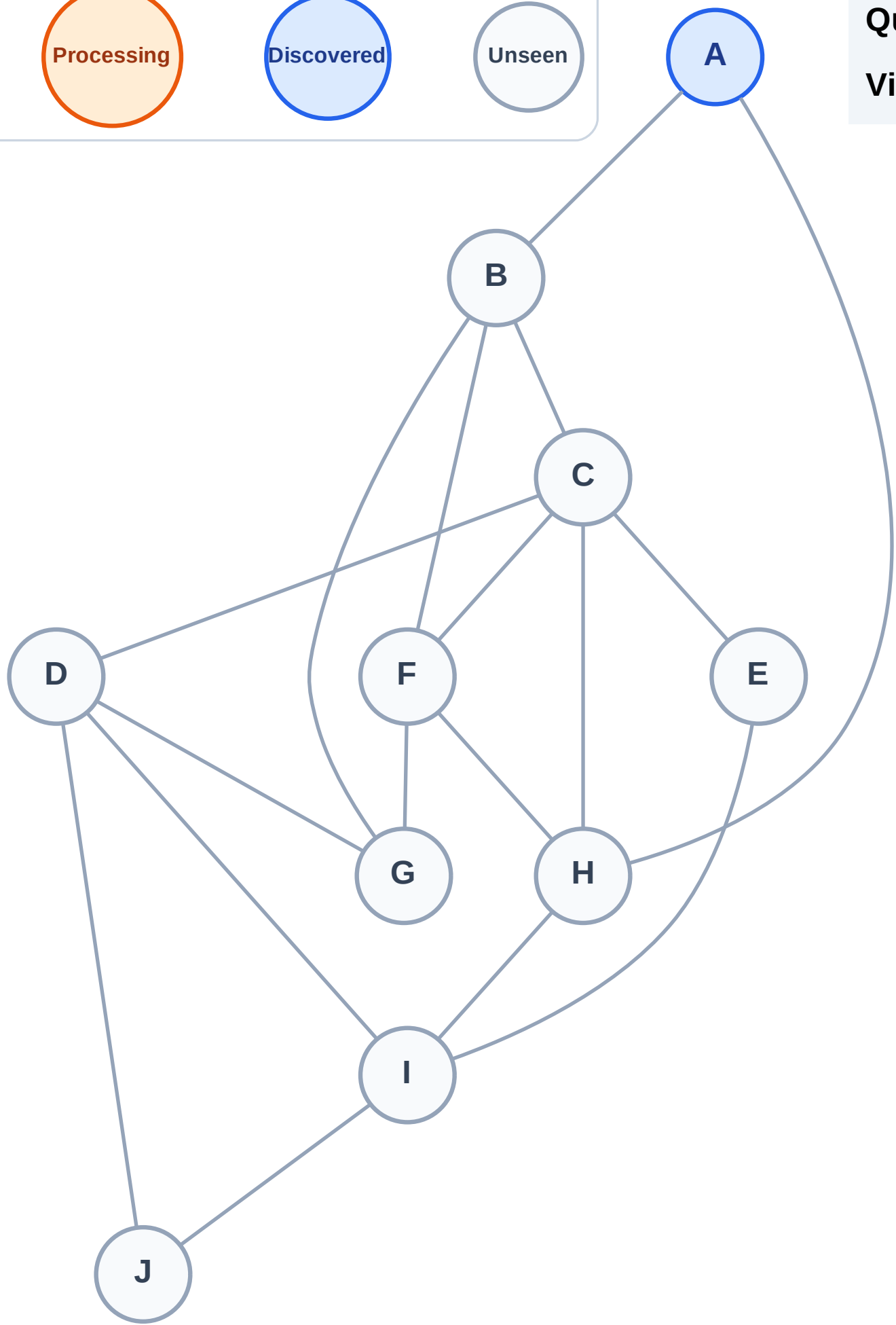
Step 1: Start at A

Legend



Queue: A

Visited: (none)



BFS Traversal

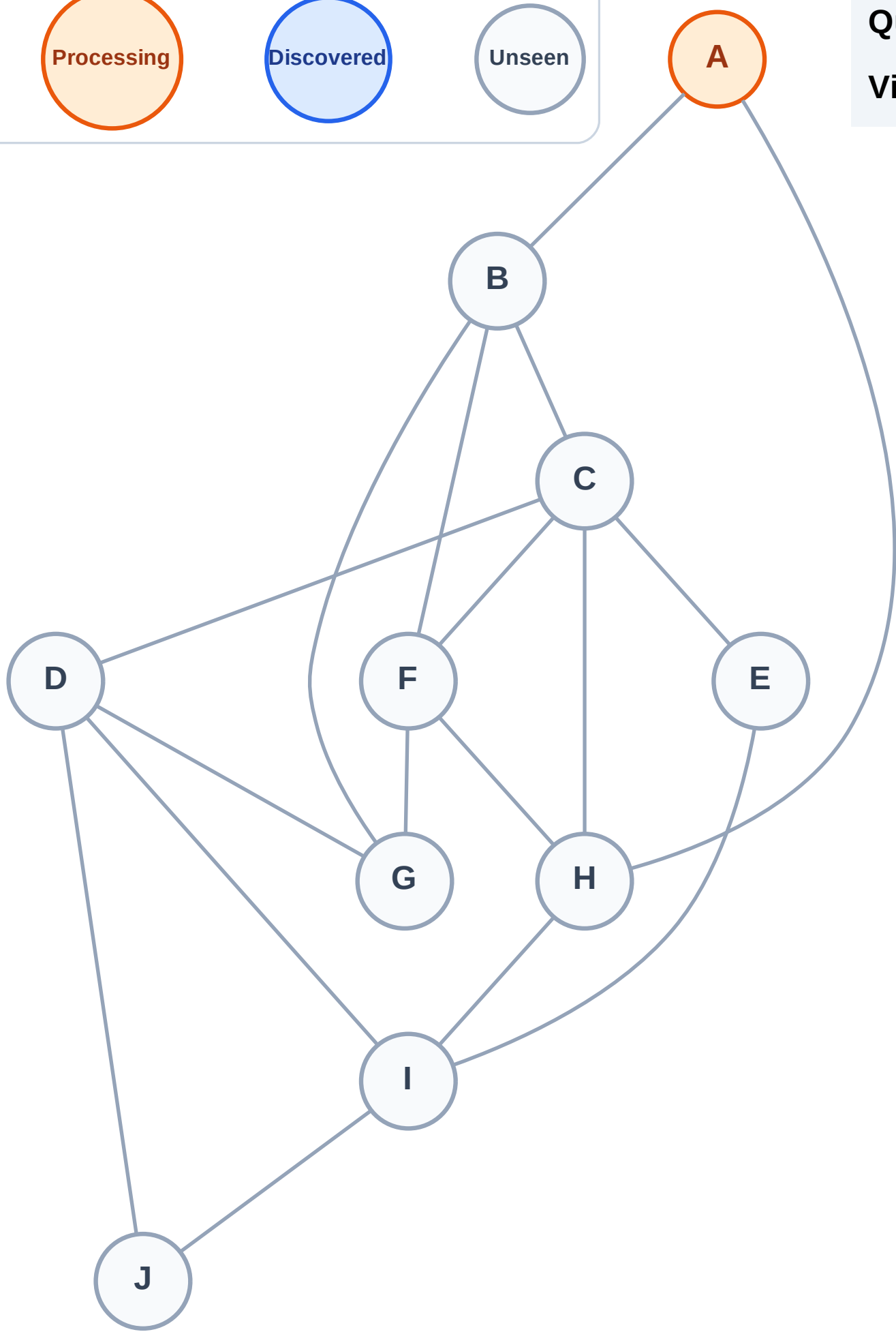
Step 2: Process node A

Legend



Queue: (empty)

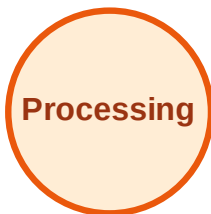
Visited: (none)



BFS Traversal

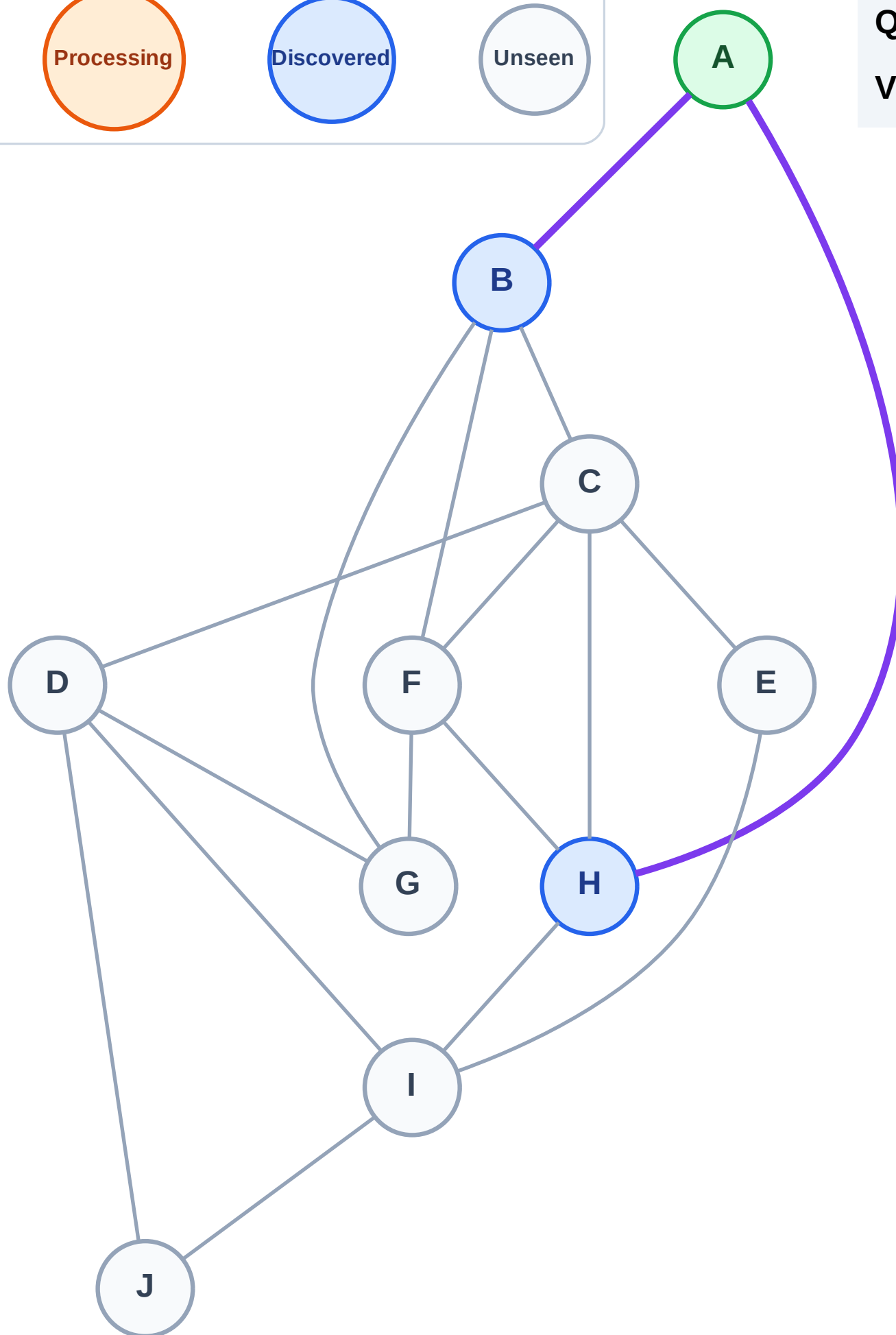
Step 3: Discover B, H from A

Legend



Queue: B → H

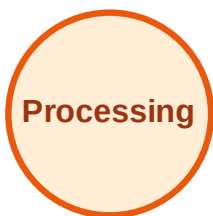
Visited: A



BFS Traversal

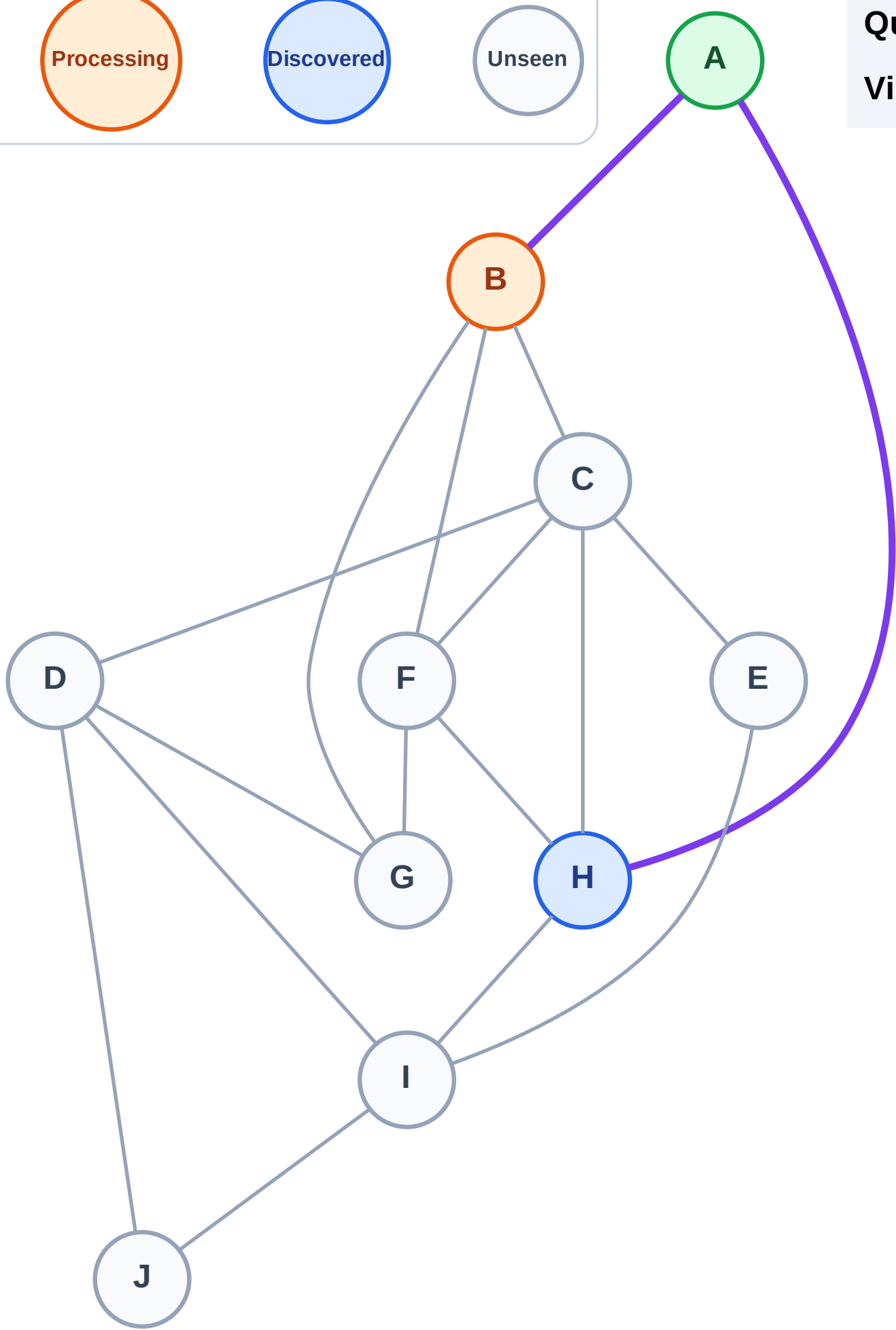
Step 4: Process node B

Legend



Queue: H

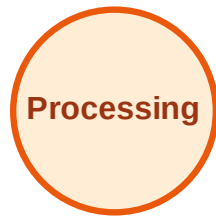
Visited: A



BFS Traversal

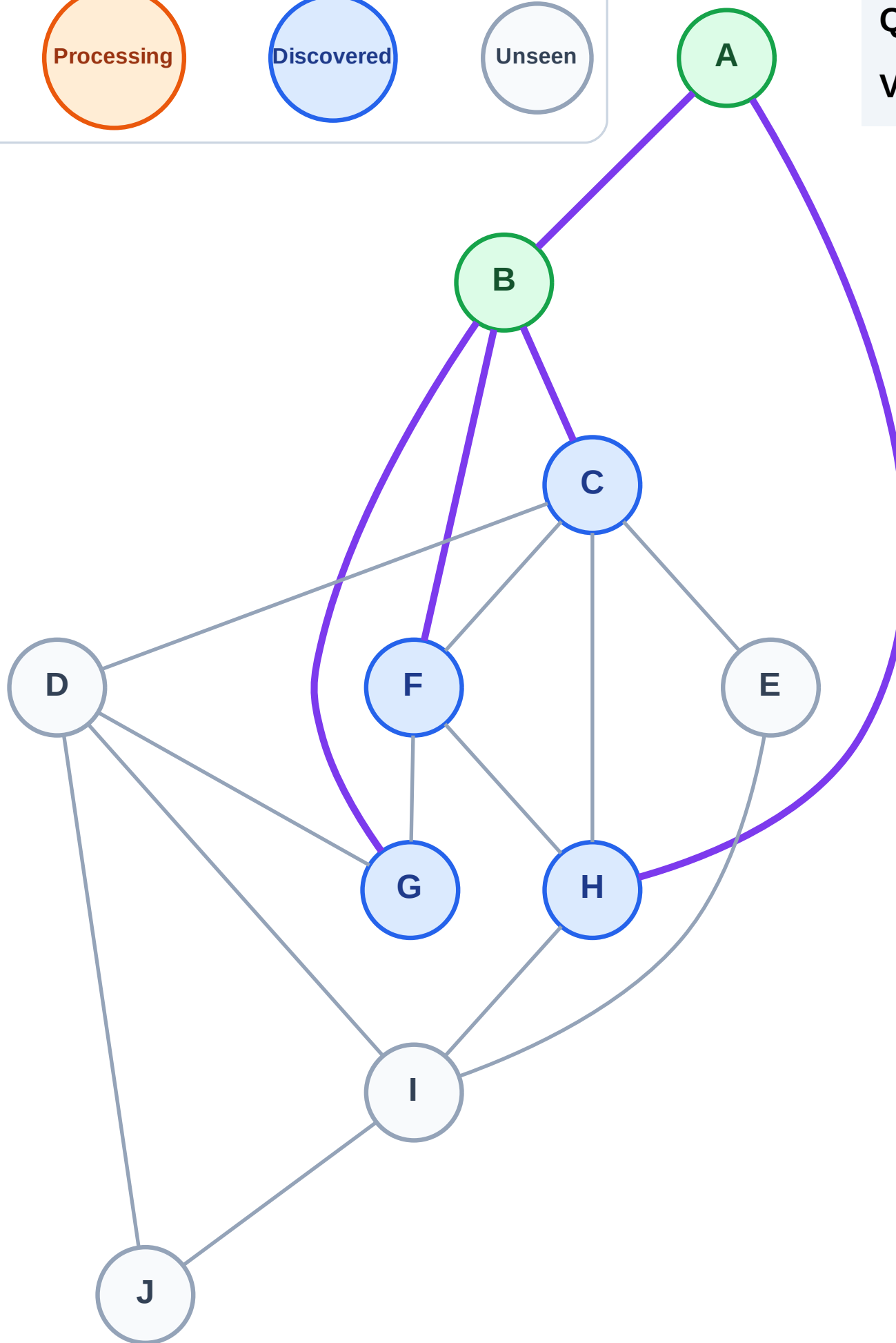
Step 5: Discover C, F, G from B

Legend



Queue: H → C → F → G

Visited: A, B



BFS Traversal

Step 6: Process node H

Legend

Complete

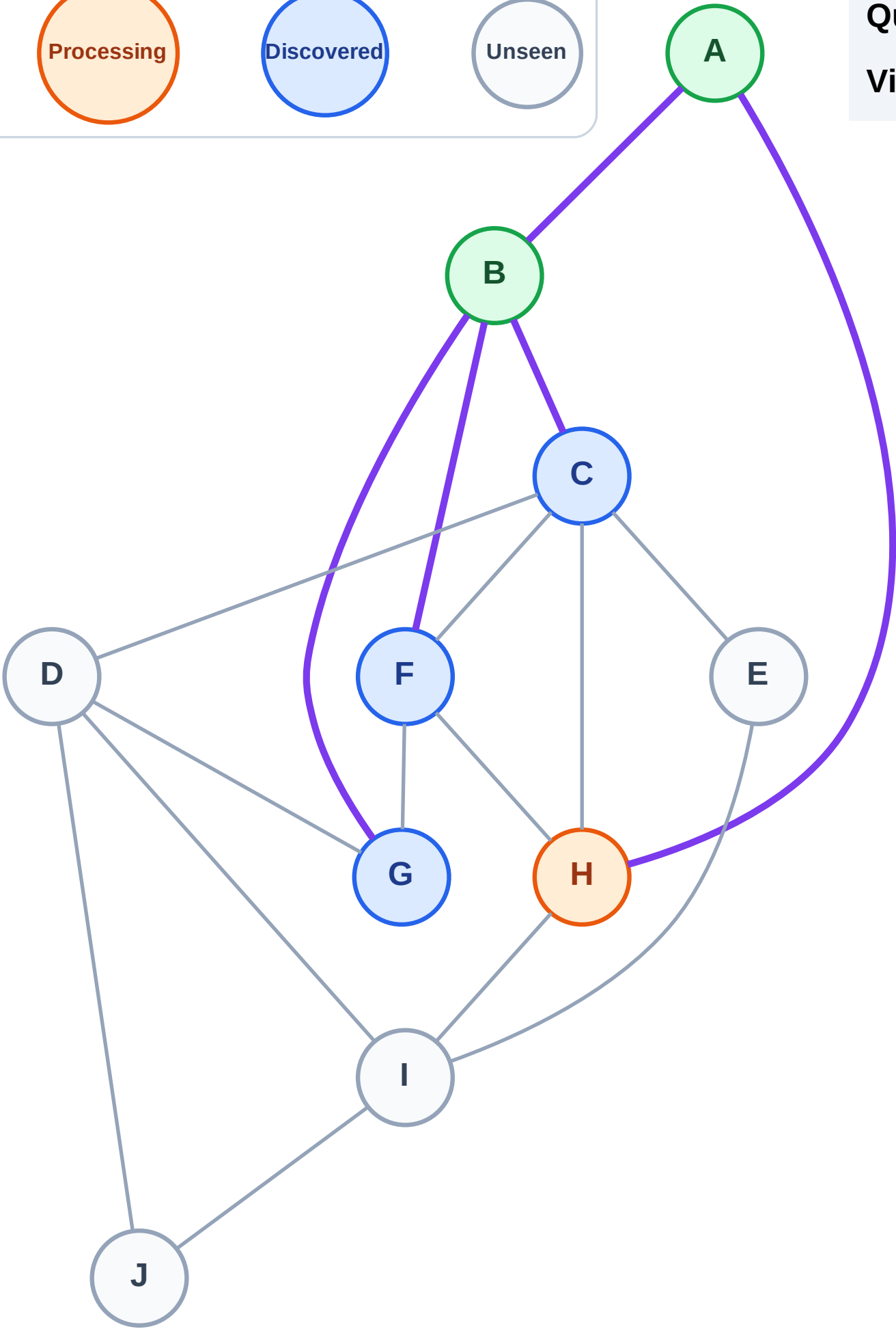
Processing

Discovered

Unseen

Queue: C → F → G

Visited: A, B



BFS Traversal

Step 7: Discover I from H

Legend

Complete

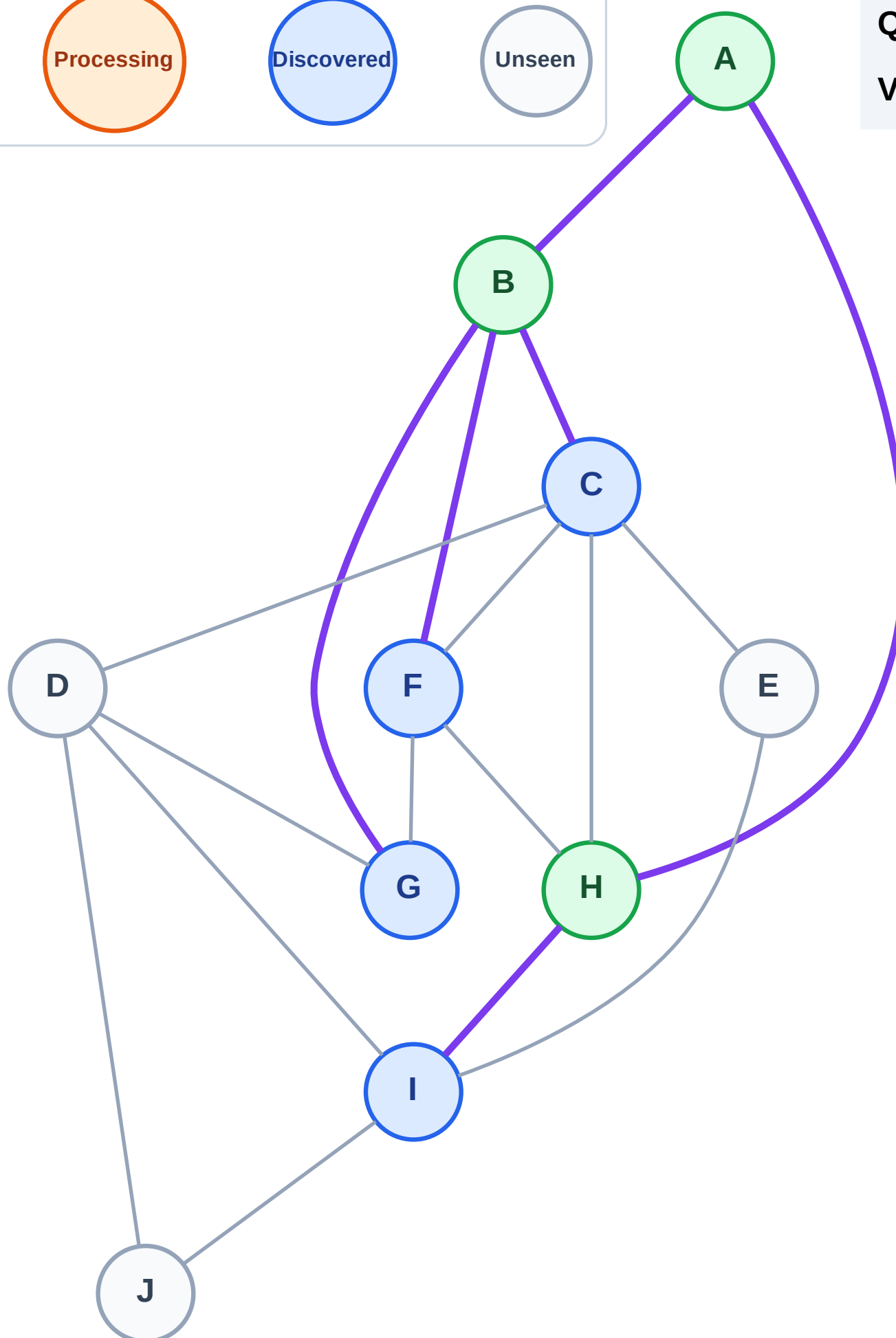
Processing

Discovered

Unseen

Queue: C → F → G → I

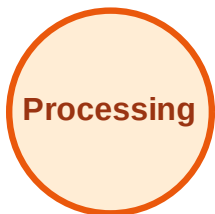
Visited: A, B, H



BFS Traversal

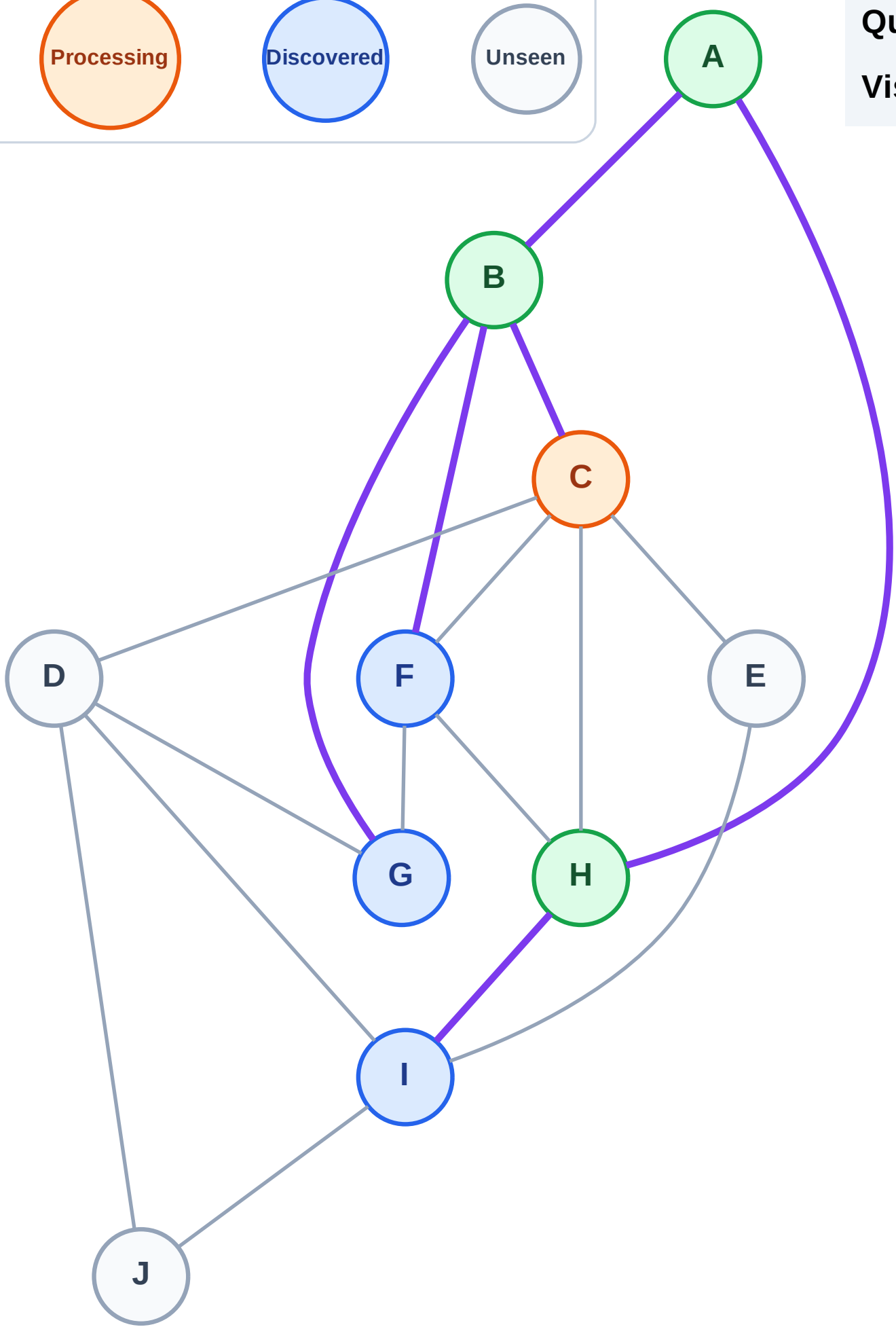
Step 8: Process node C

Legend



Queue: F → G → I

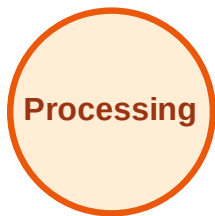
Visited: A, B, H



BFS Traversal

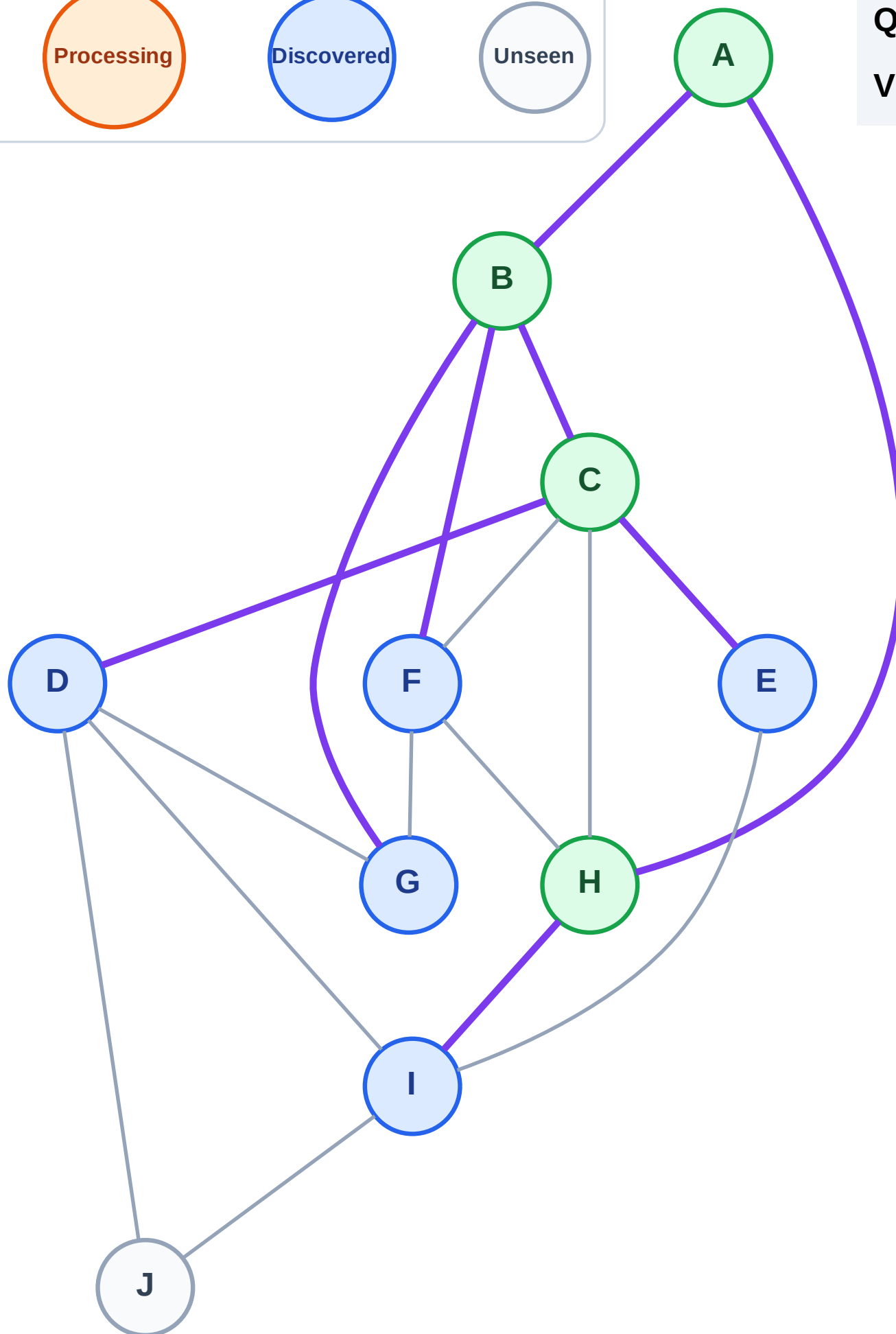
Step 9: Discover D, E from C

Legend



Queue: F → G → I → D → E

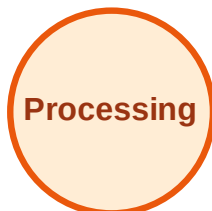
Visited: A, B, C, H



BFS Traversal

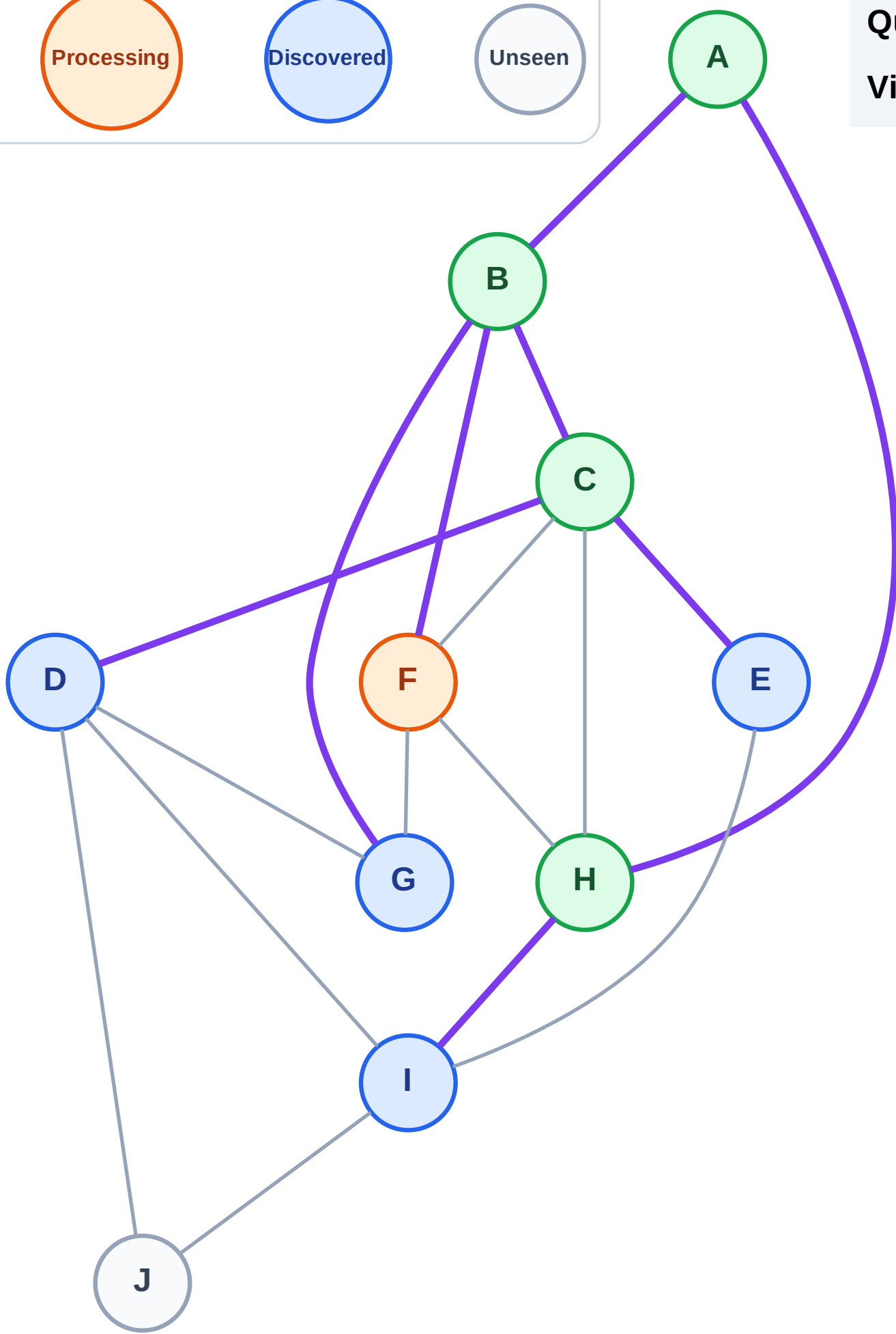
Step 10: Process node F

Legend



Queue: G → I → D → E

Visited: A, B, C, H



BFS Traversal

Step 11: No new neighbors from F

Legend

Complete

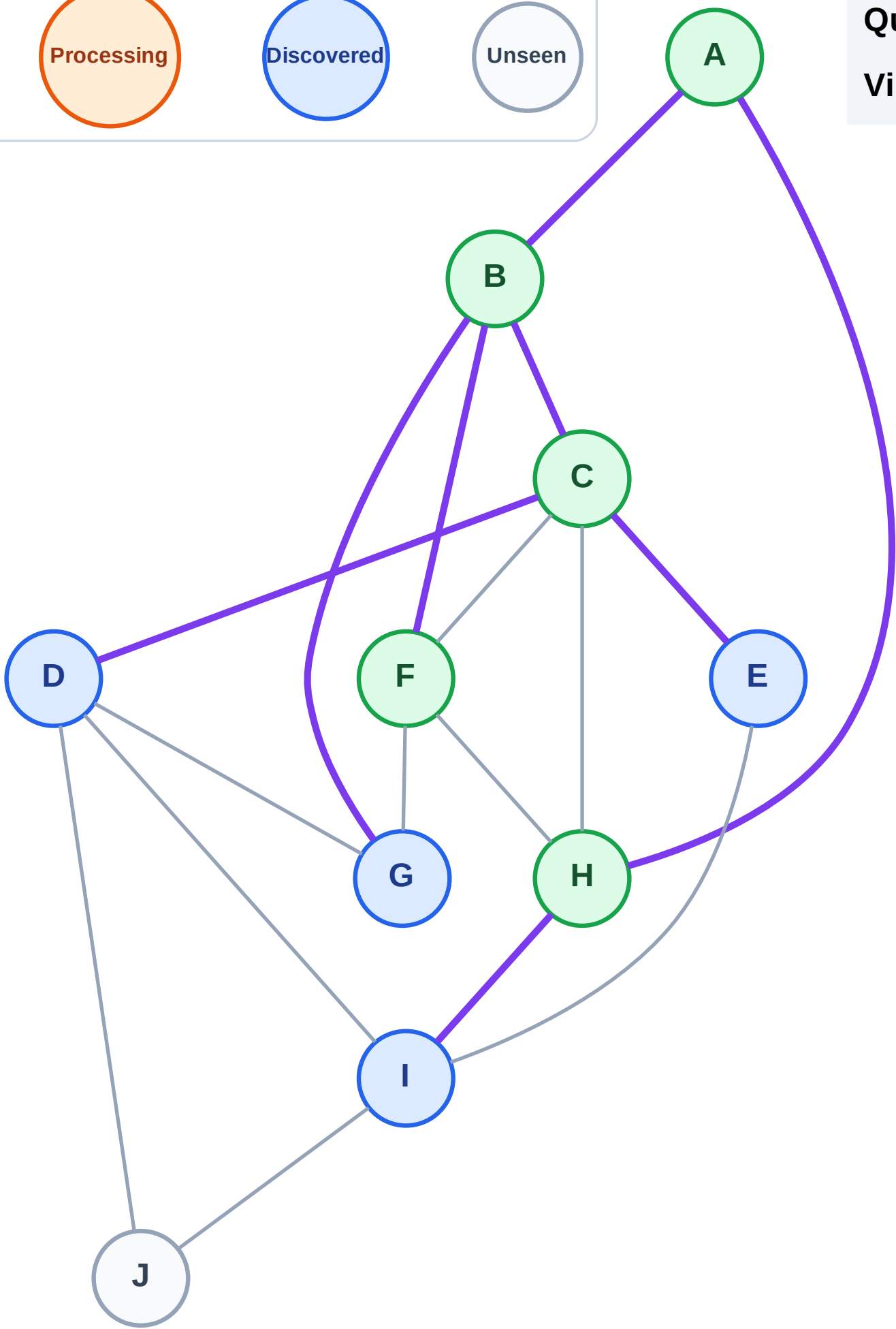
Processing

Discovered

Unseen

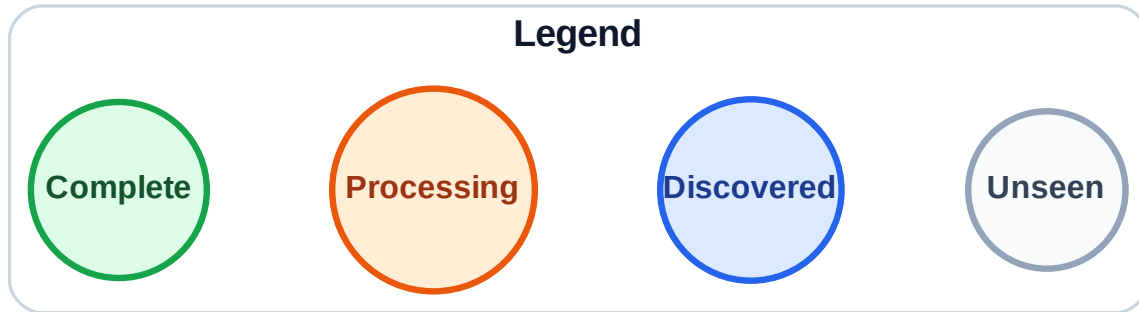
Queue: G → I → D → E

Visited: A, B, C, F, H

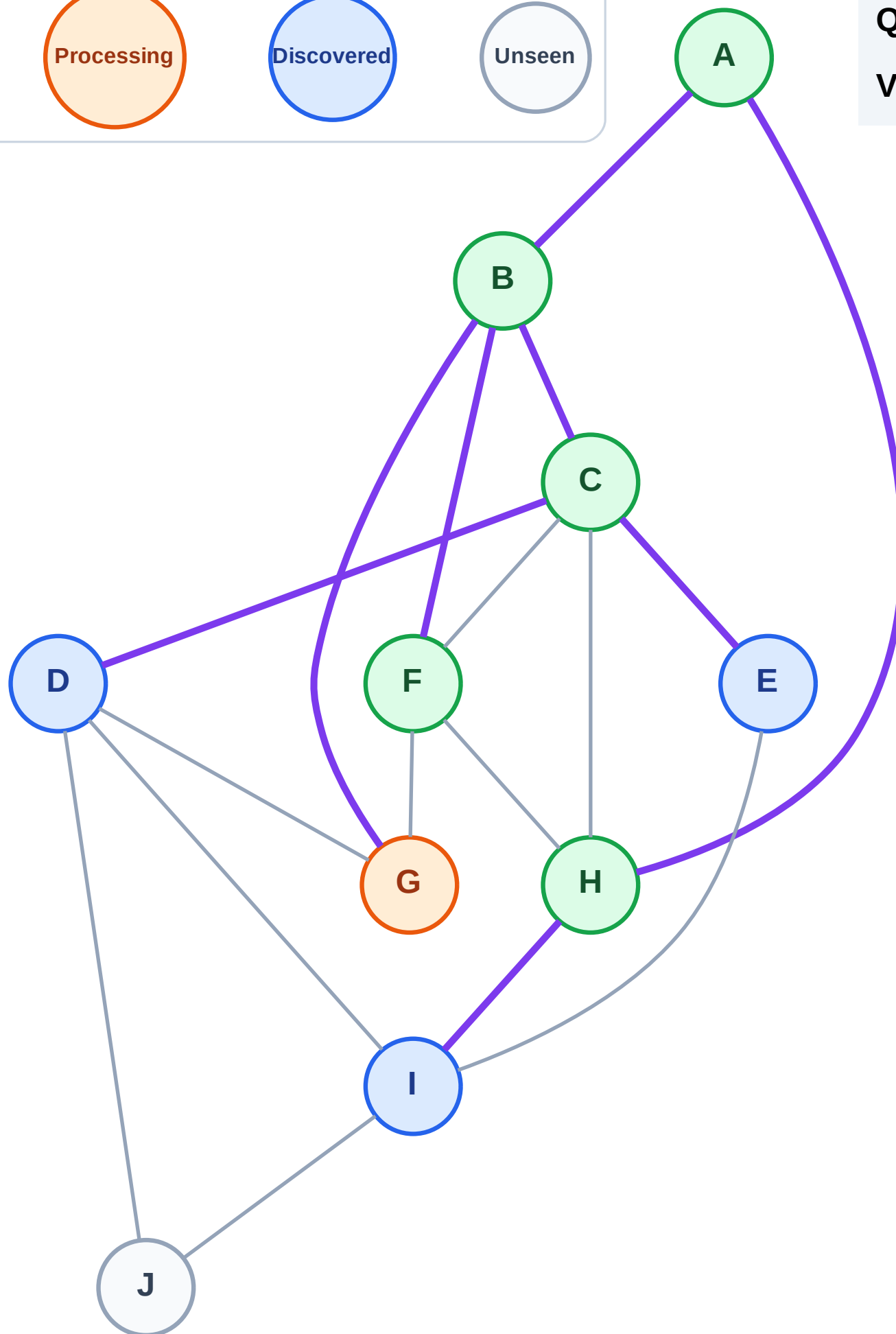


BFS Traversal

Step 12: Process node G



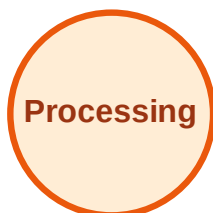
Queue: I → D → E
Visited: A, B, C, F, H



BFS Traversal

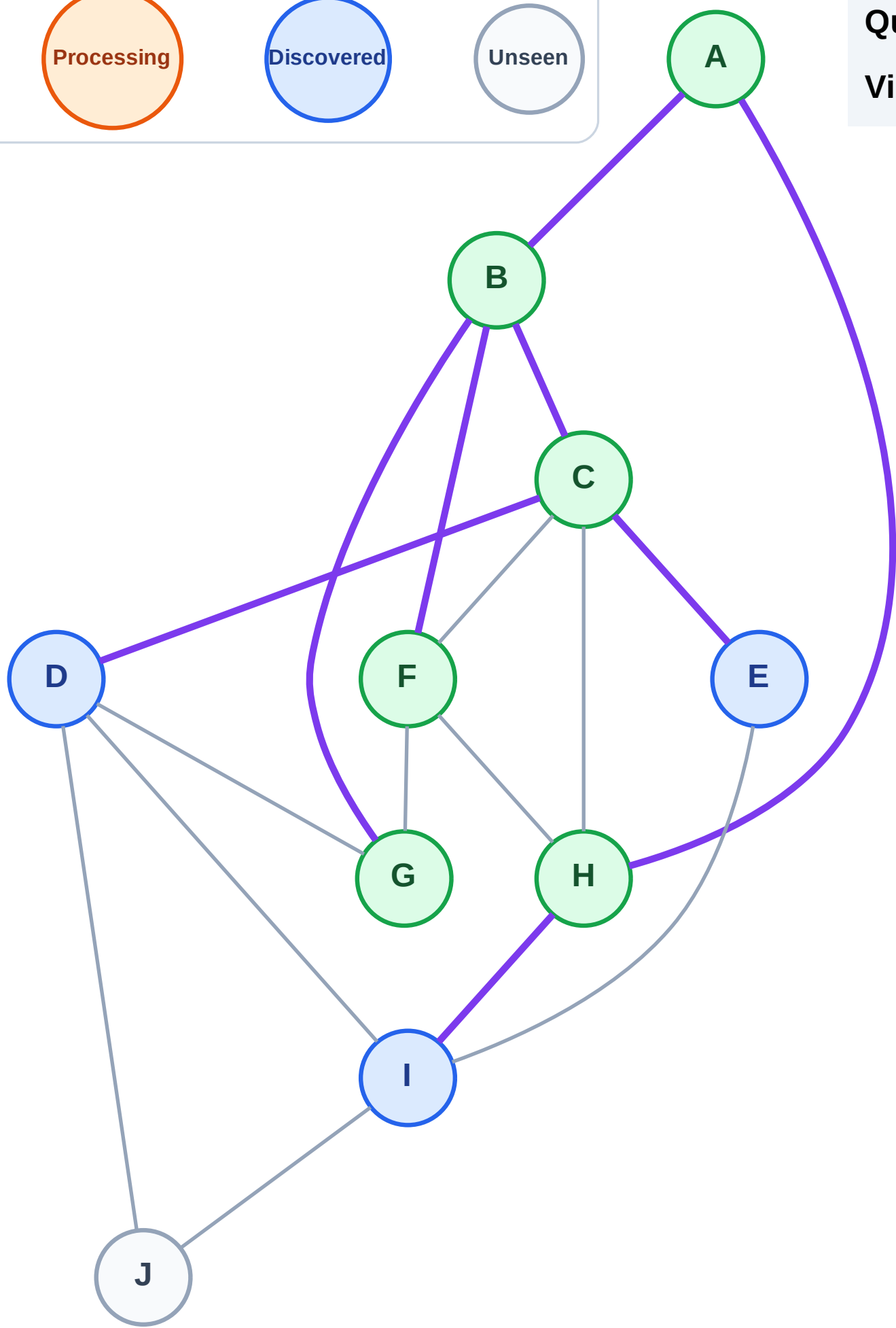
Step 13: No new neighbors from G

Legend



Queue: I → D → E

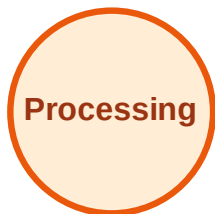
Visited: A, B, C, F, G, H



BFS Traversal

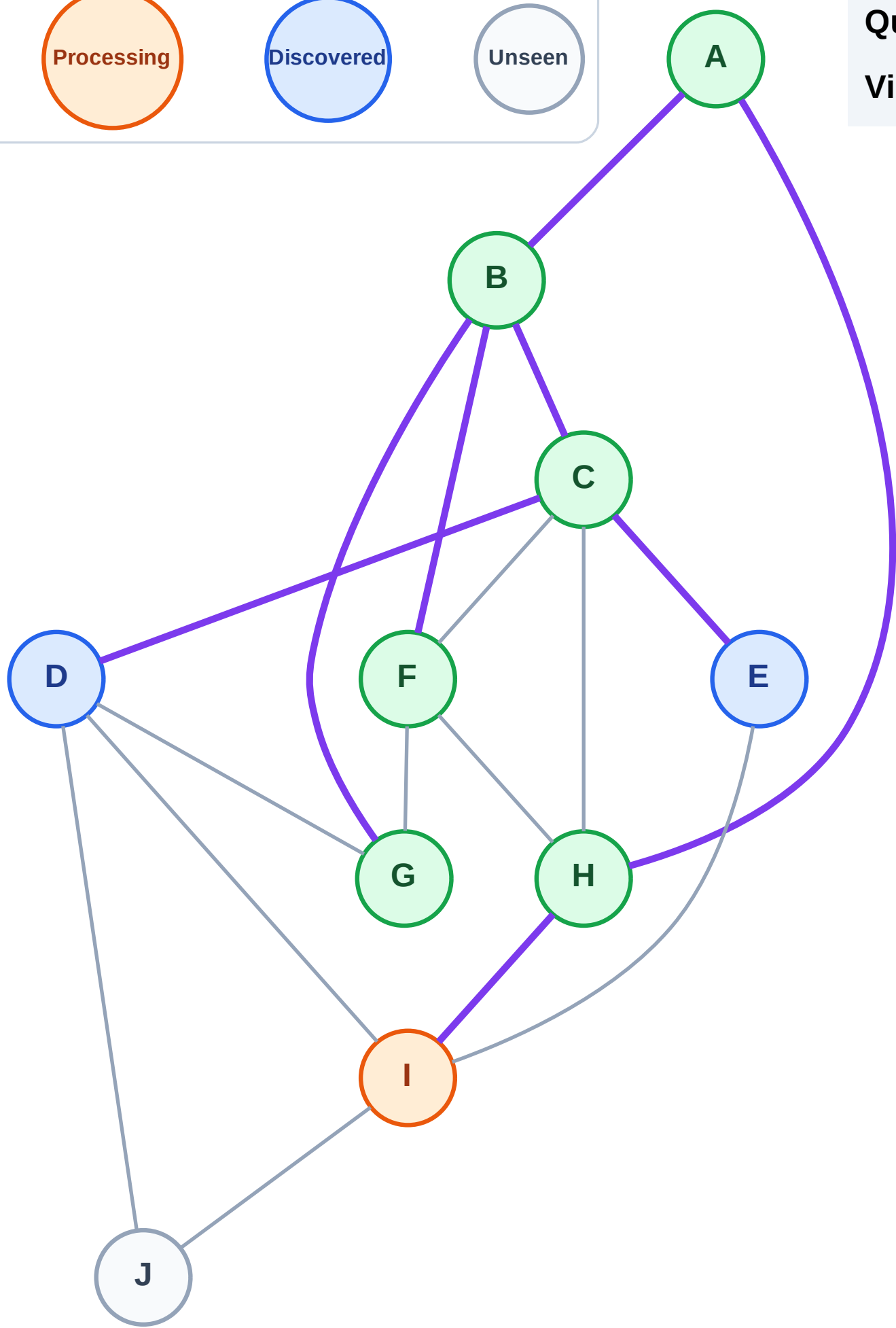
Step 14: Process node I

Legend



Queue: D → E

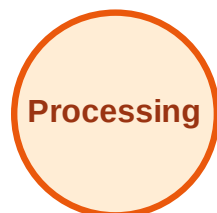
Visited: A, B, C, F, G, H



BFS Traversal

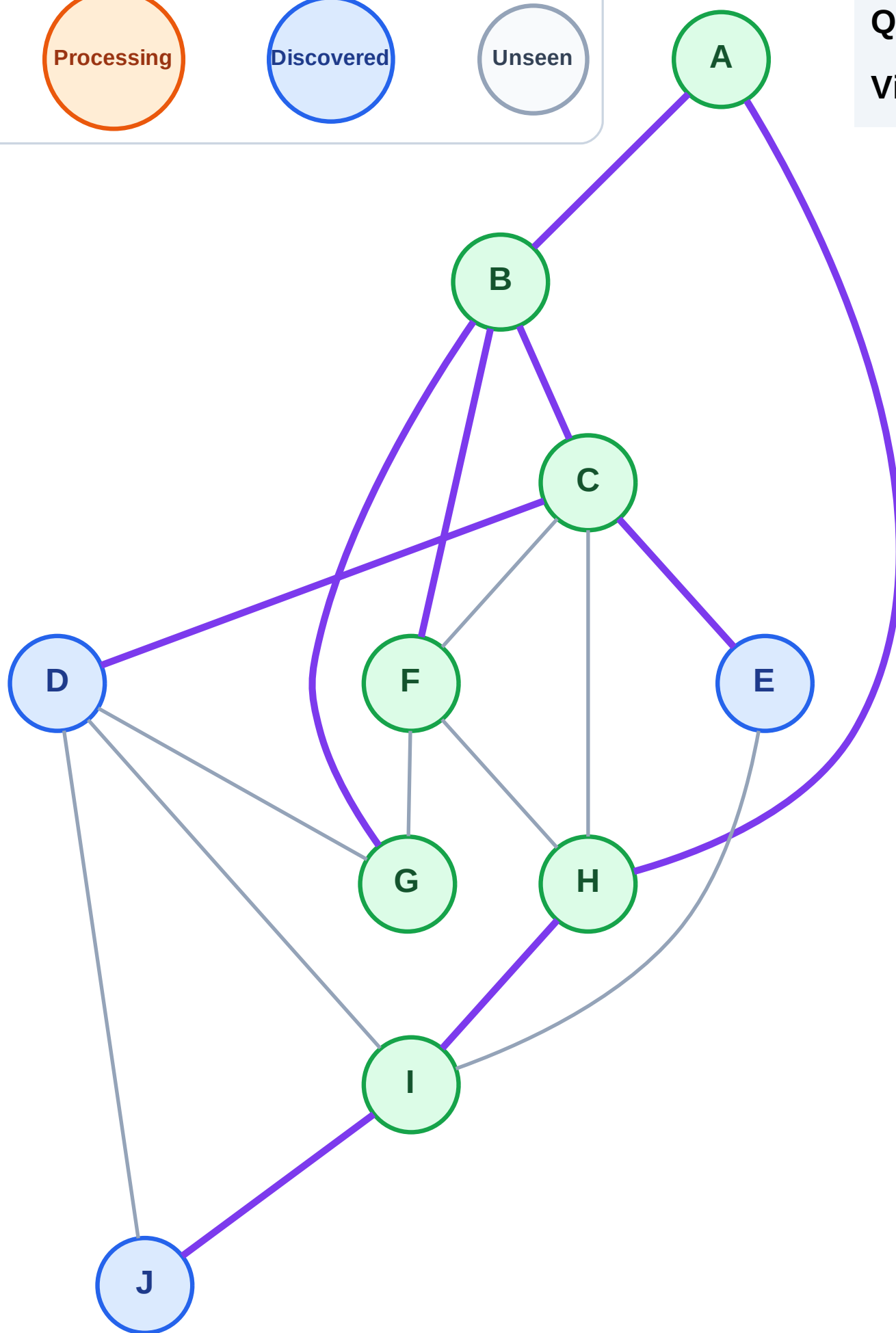
Step 15: Discover J from I

Legend



Queue: D → E → J

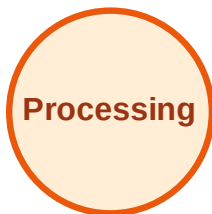
Visited: A, B, C, F, G, H, I



BFS Traversal

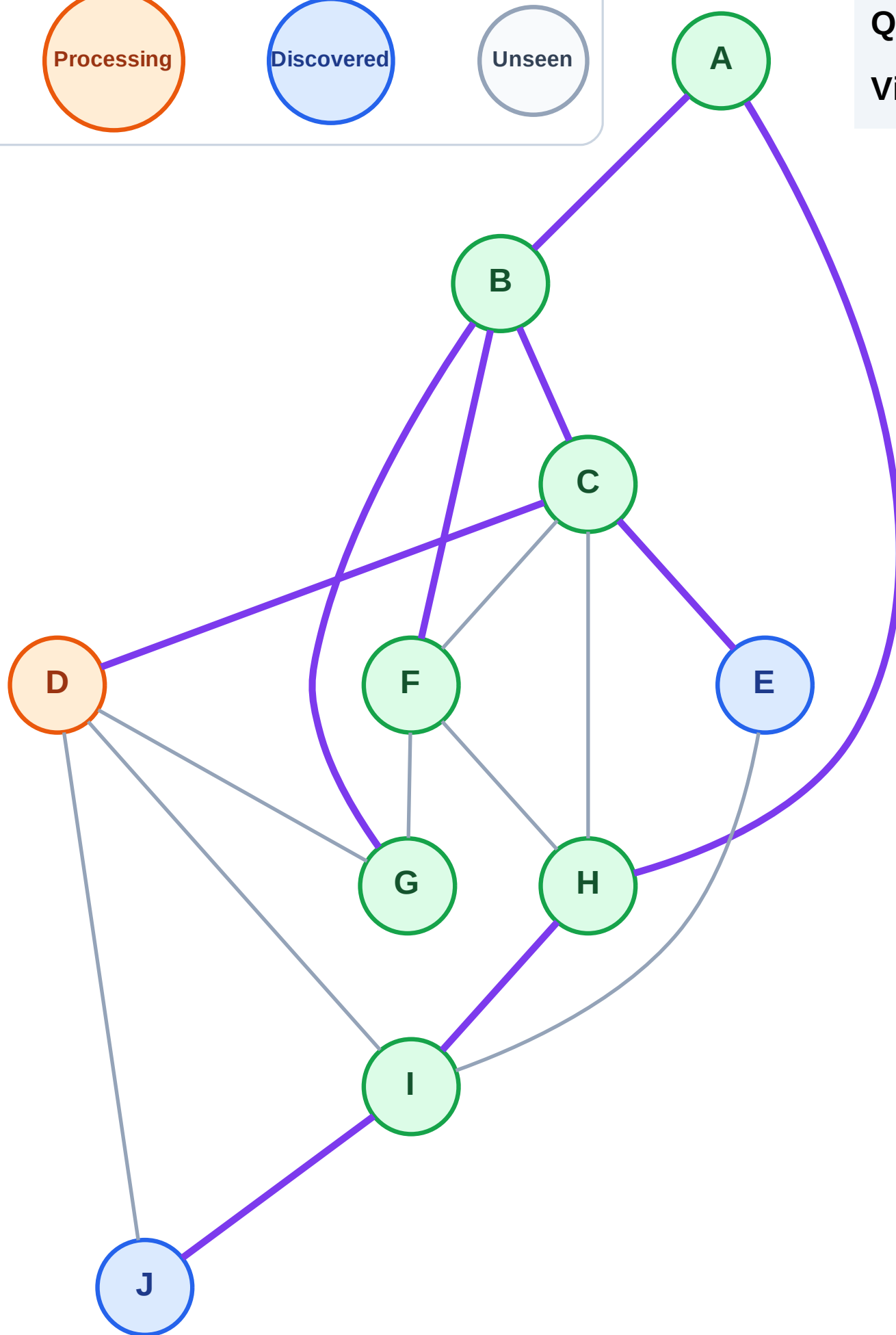
Step 16: Process node D

Legend



Queue: E → J

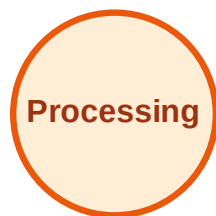
Visited: A, B, C, F, G, H, I



BFS Traversal

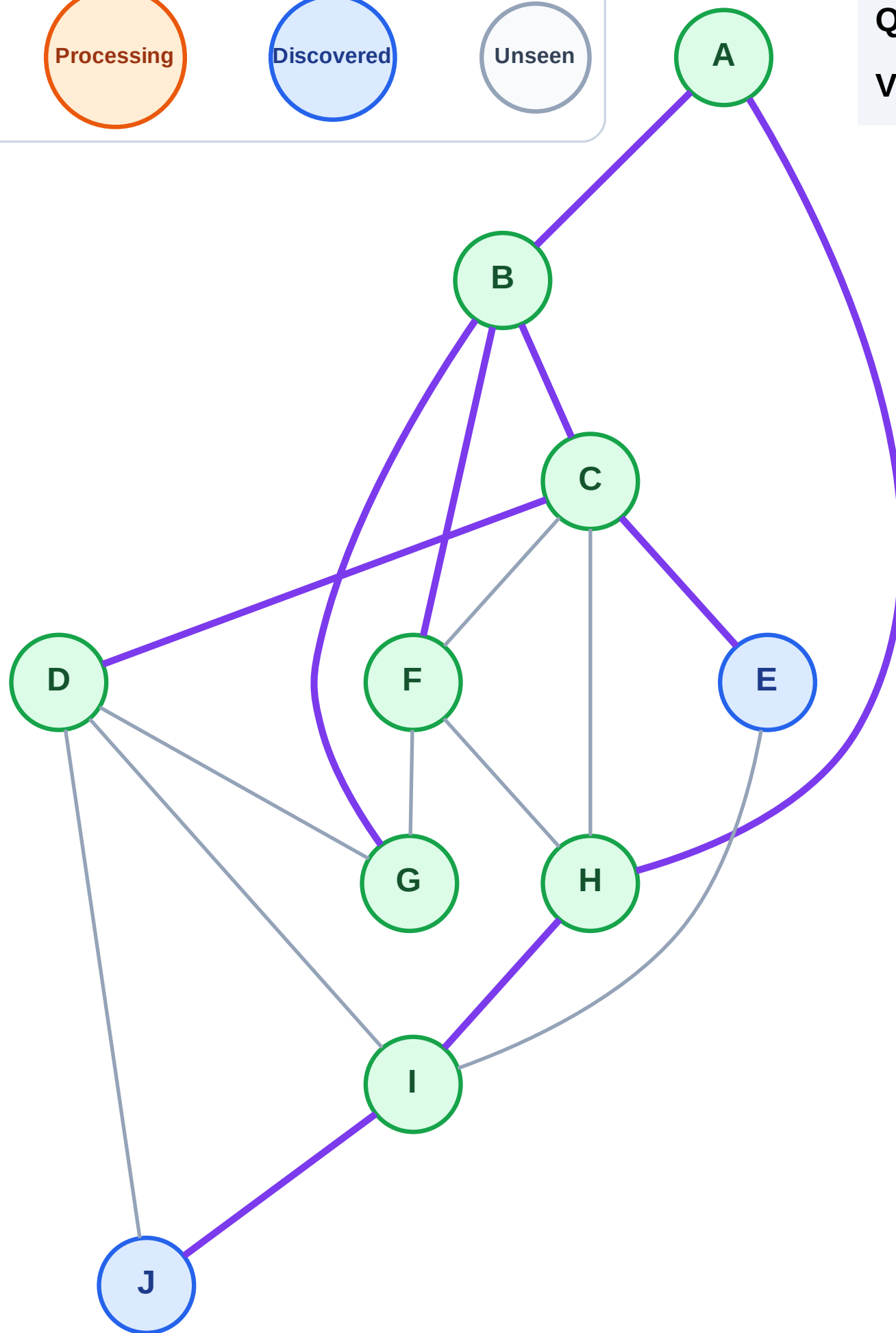
Step 17: No new neighbors from D

Legend



Queue: E → J

Visited: A, B, C, D, F, G, H, I



BFS Traversal

Step 18: Process node E

Legend

Complete

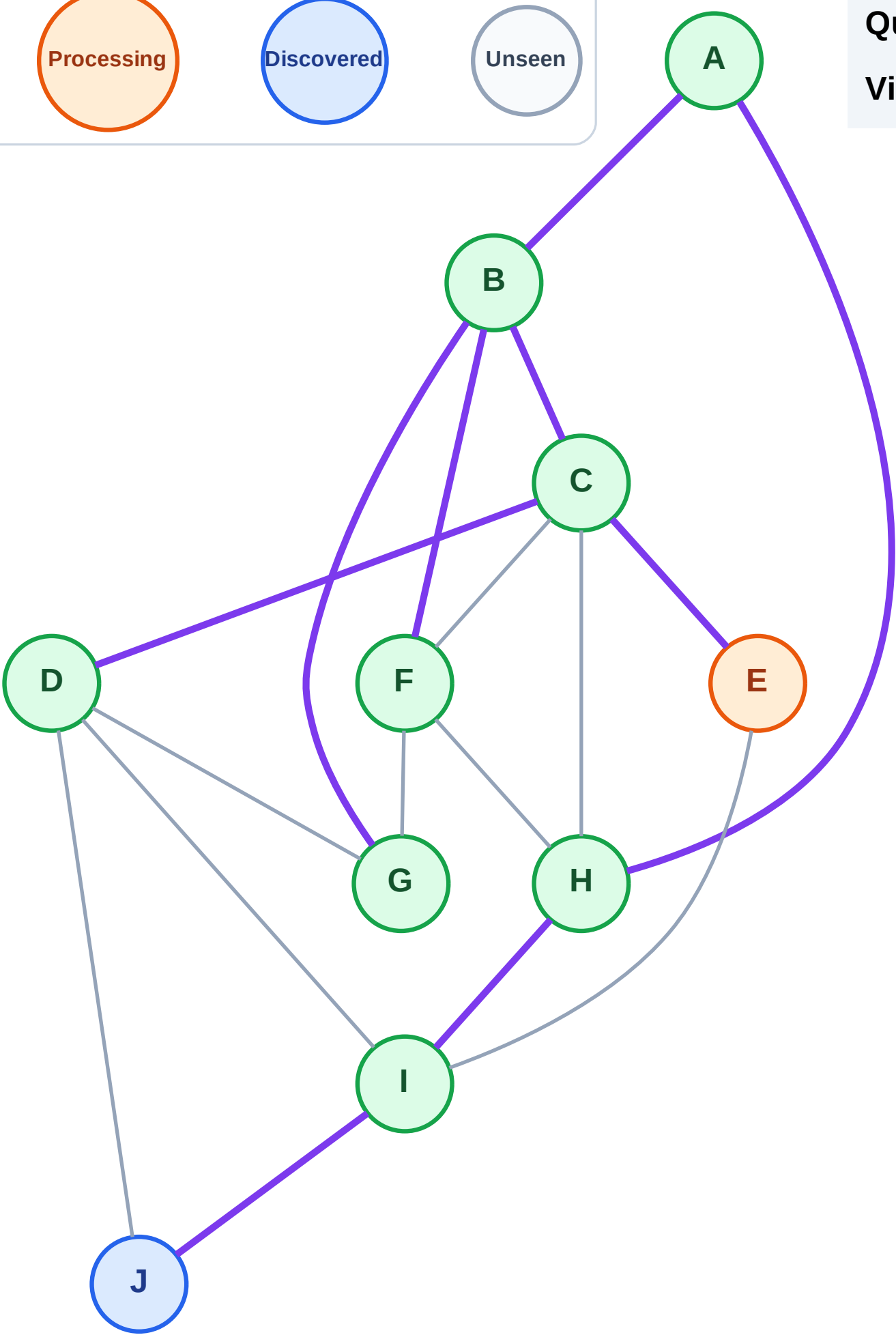
Processing

Discovered

Unseen

Queue: J

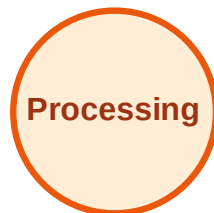
Visited: A, B, C, D, F, G, H, I



BFS Traversal

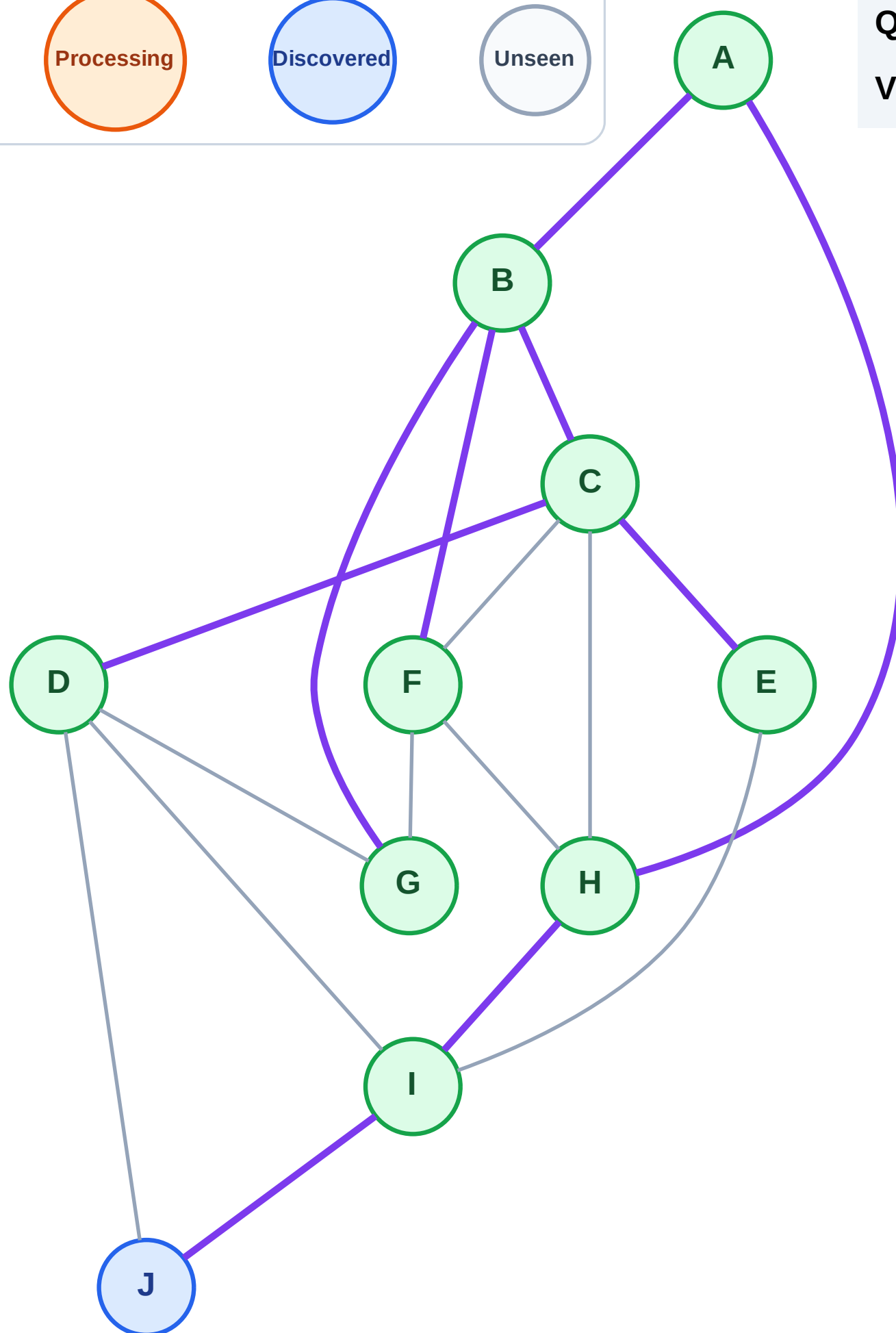
Step 19: No new neighbors from E

Legend



Queue: J

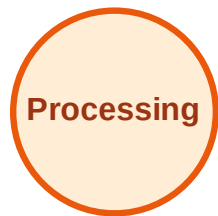
Visited: A, B, C, D, E, F, G, H, I



BFS Traversal

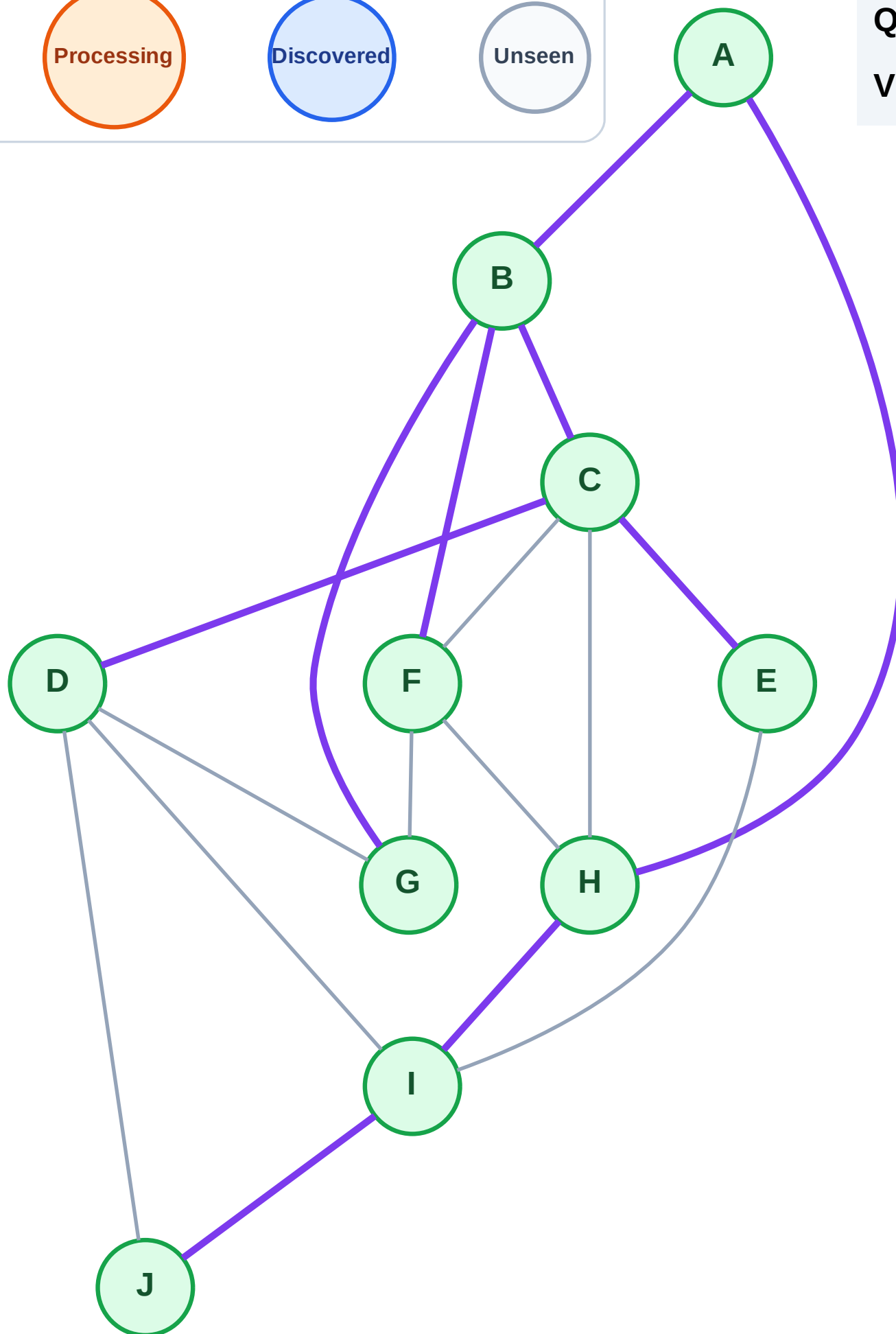
Step 21: No new neighbors from J

Legend



Queue: (empty)

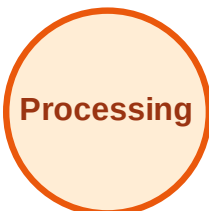
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

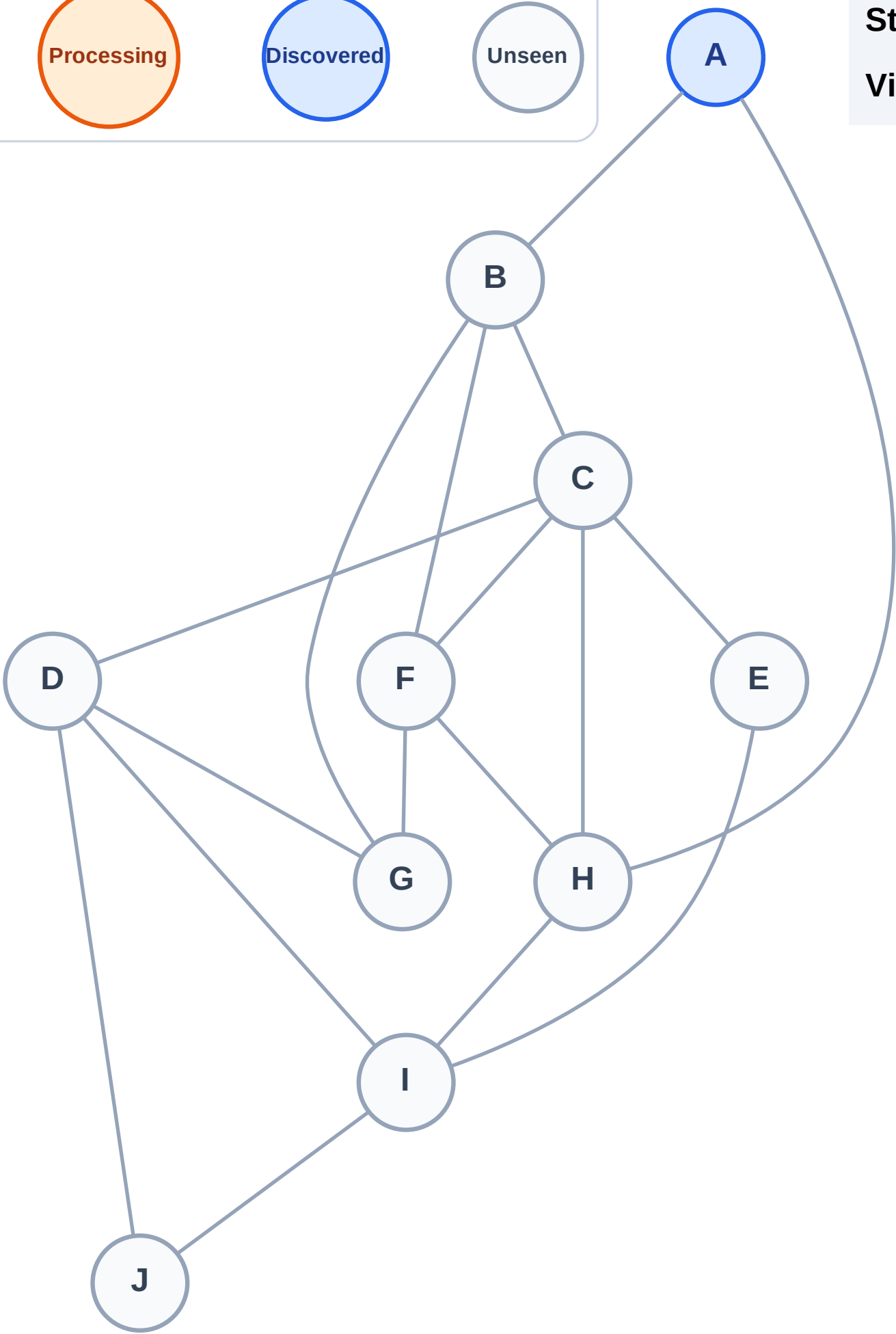
Step 1: Start at A

Legend



Stack (top at right): A

Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

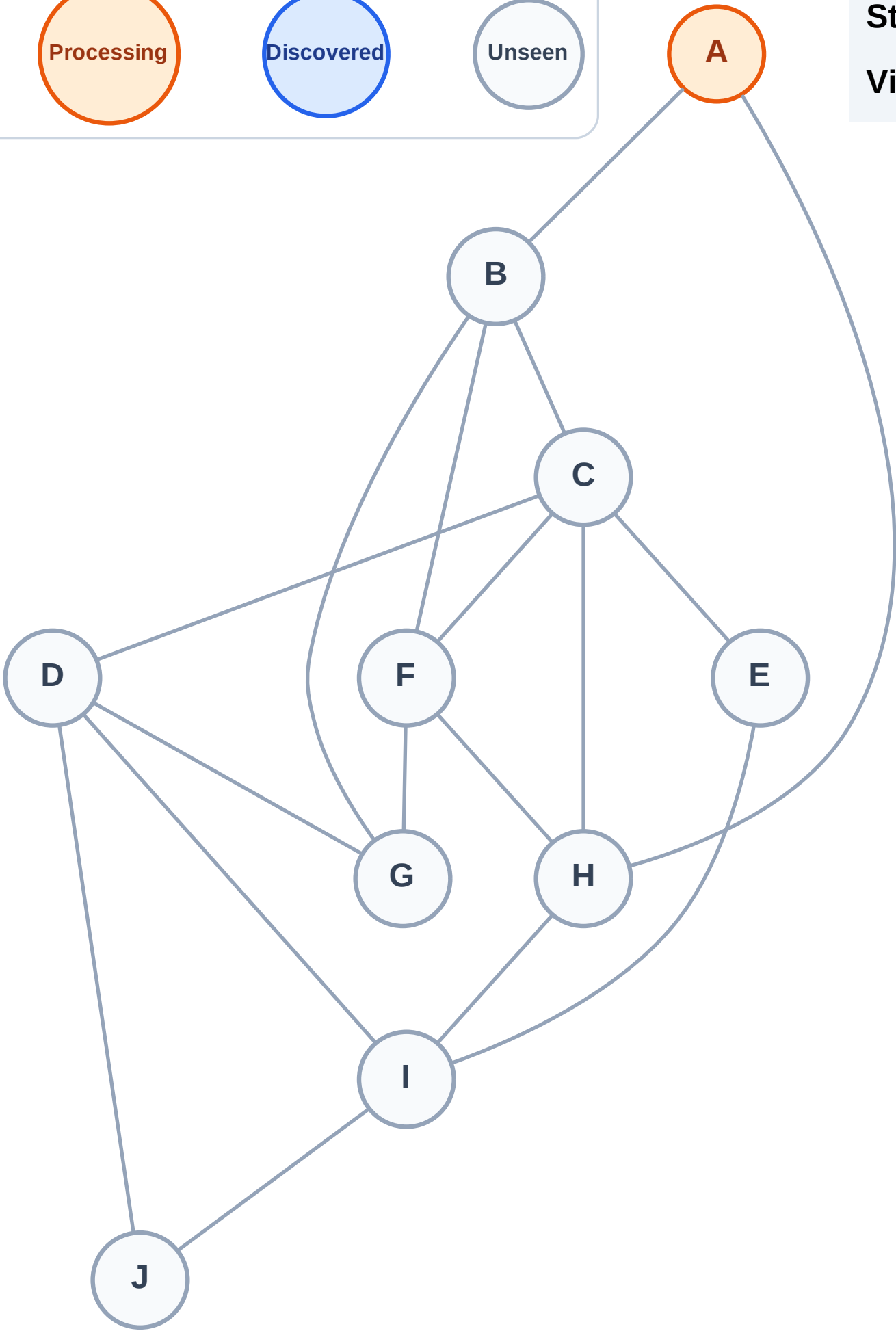
Complete

Processing

Discovered

Unseen

Stack (top at right): (empty)
Visited: (none)



DFS Traversal

Step 3: Discover H, B from A

Legend

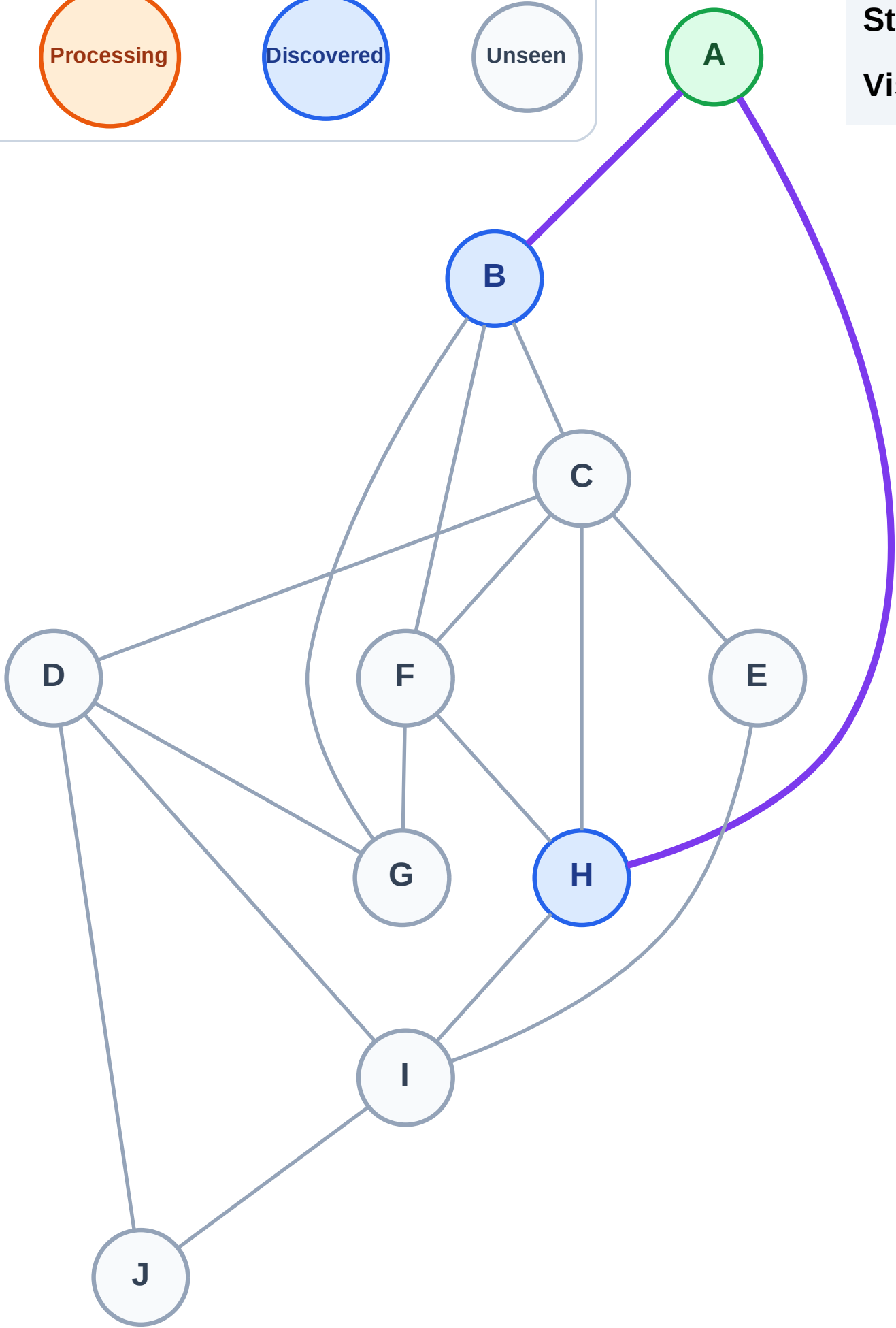
Complete

Processing

Discovered

Unseen

Stack (top at right): H | B
Visited: A



DFS Traversal

Step 4: Process node B

Legend

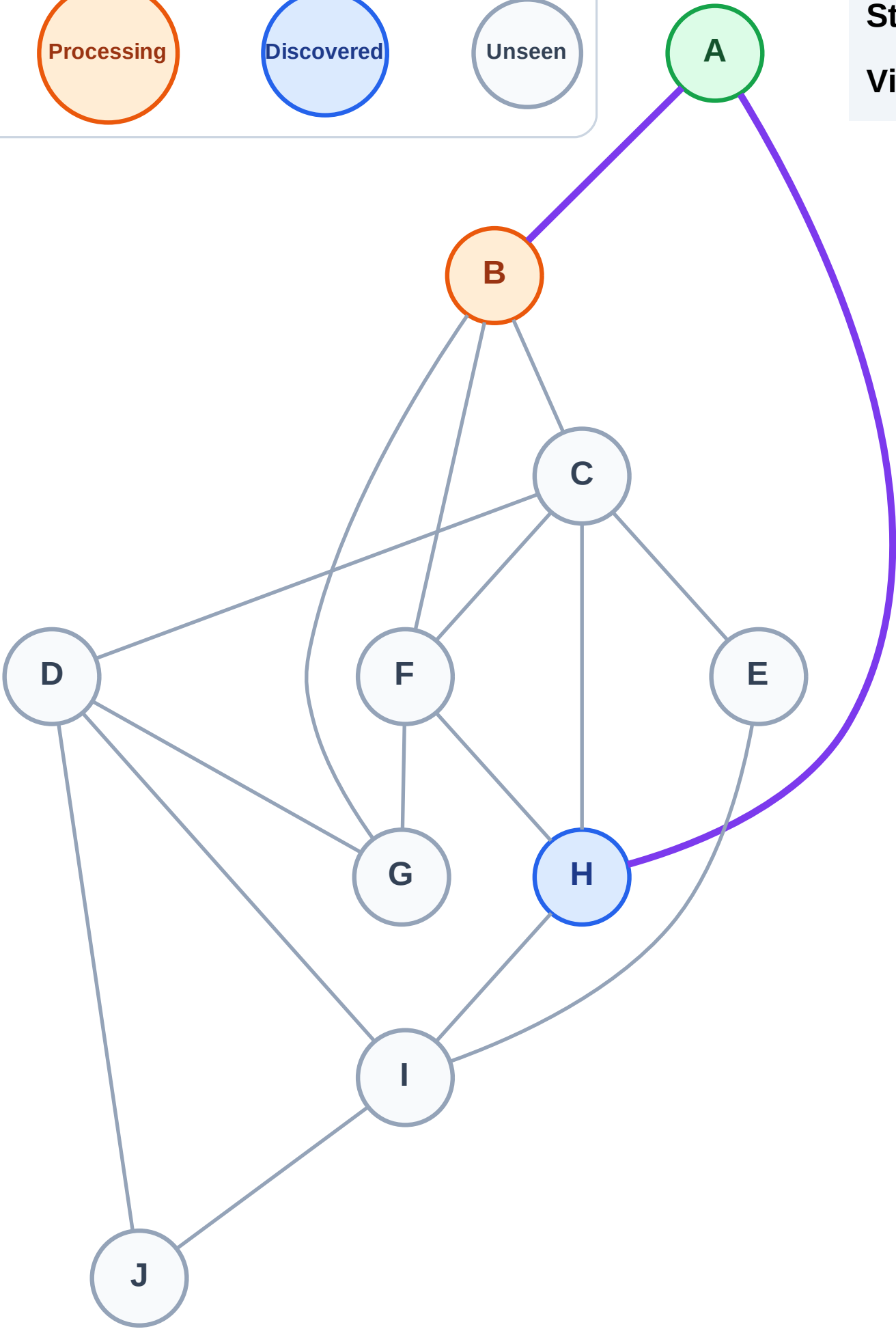
Complete

Processing

Discovered

Unseen

Stack (top at right): H
Visited: A



DFS Traversal

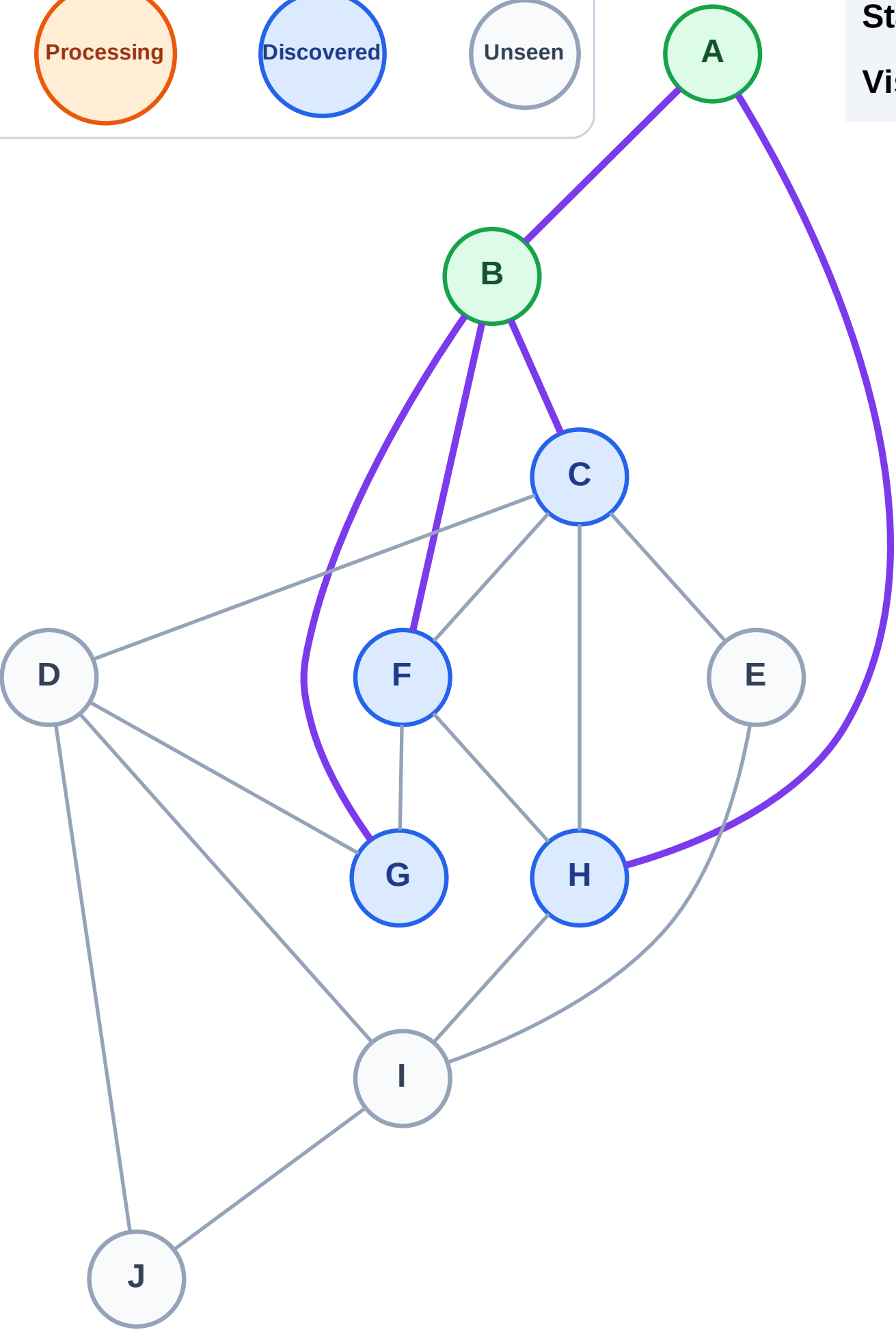
Step 5: Discover G, F, C from B

Legend



Stack (top at right): H | G | F | C

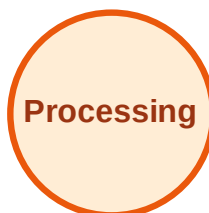
Visited: A, B



DFS Traversal

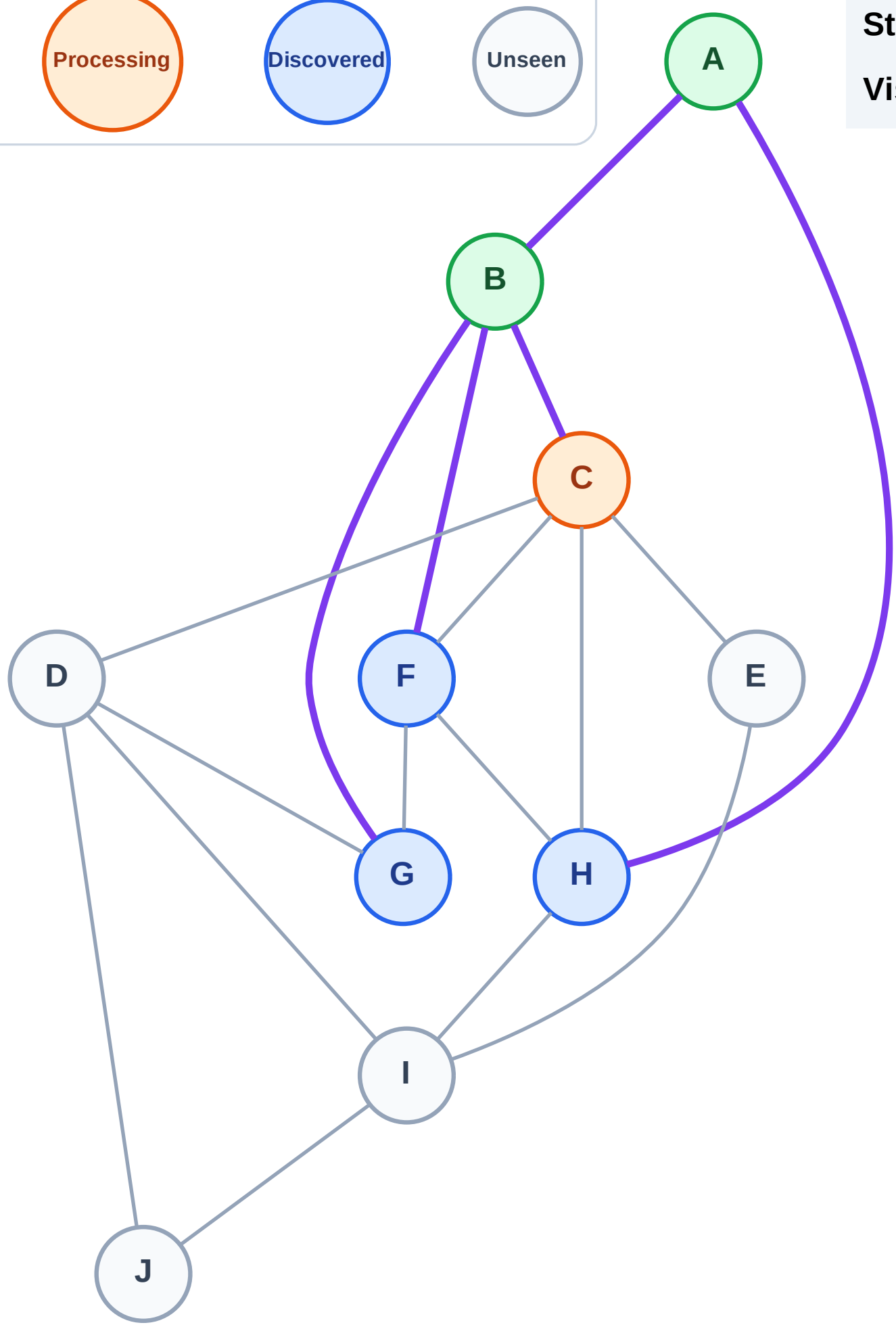
Step 6: Process node C

Legend



Stack (top at right): H | G | F

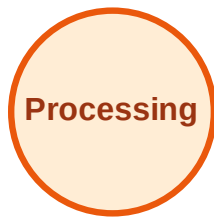
Visited: A, B



DFS Traversal

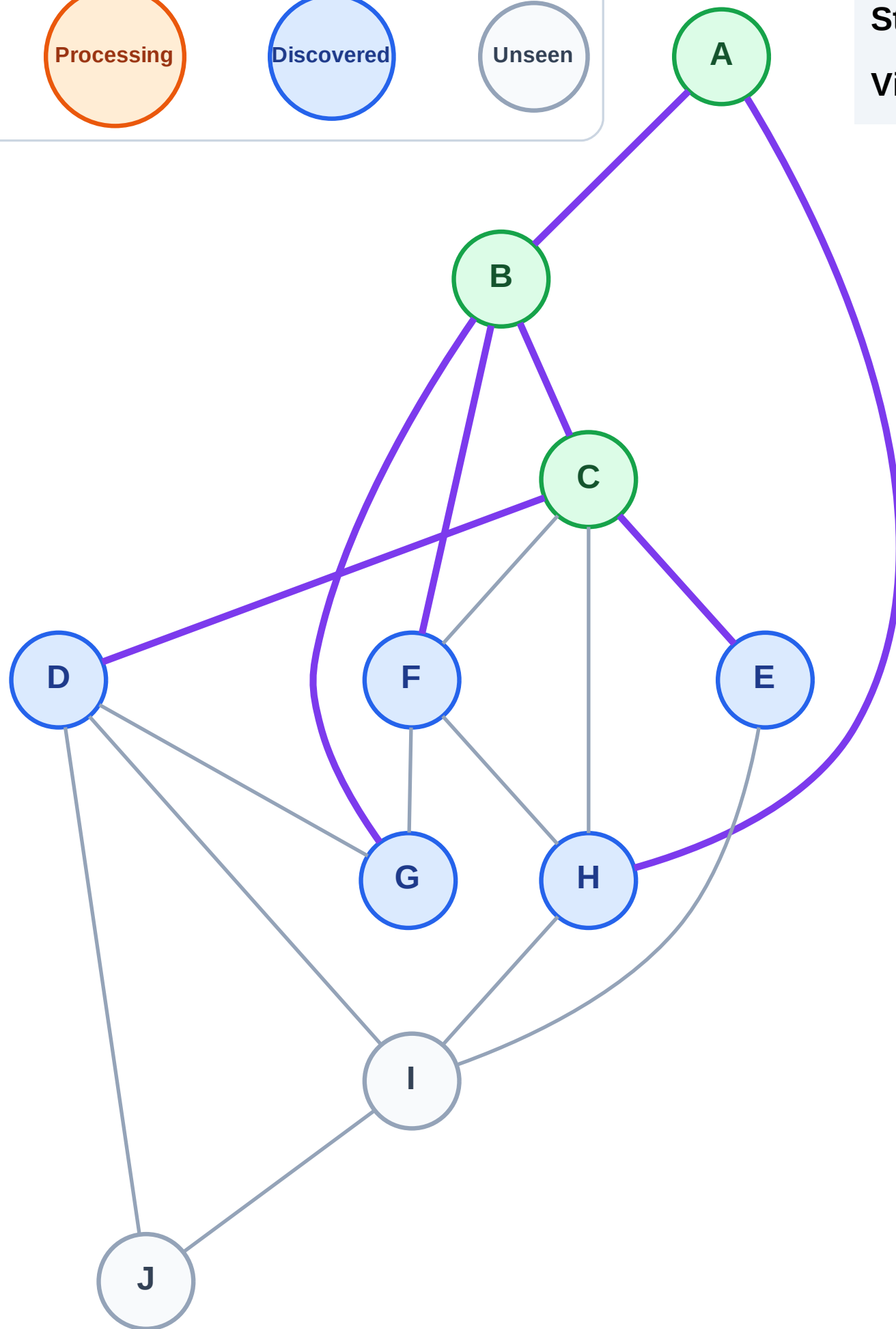
Step 7: Discover E, D from C

Legend



Stack (top at right): H | G | F | E | D

Visited: A, B, C



DFS Traversal

Step 8: Process node D

Legend

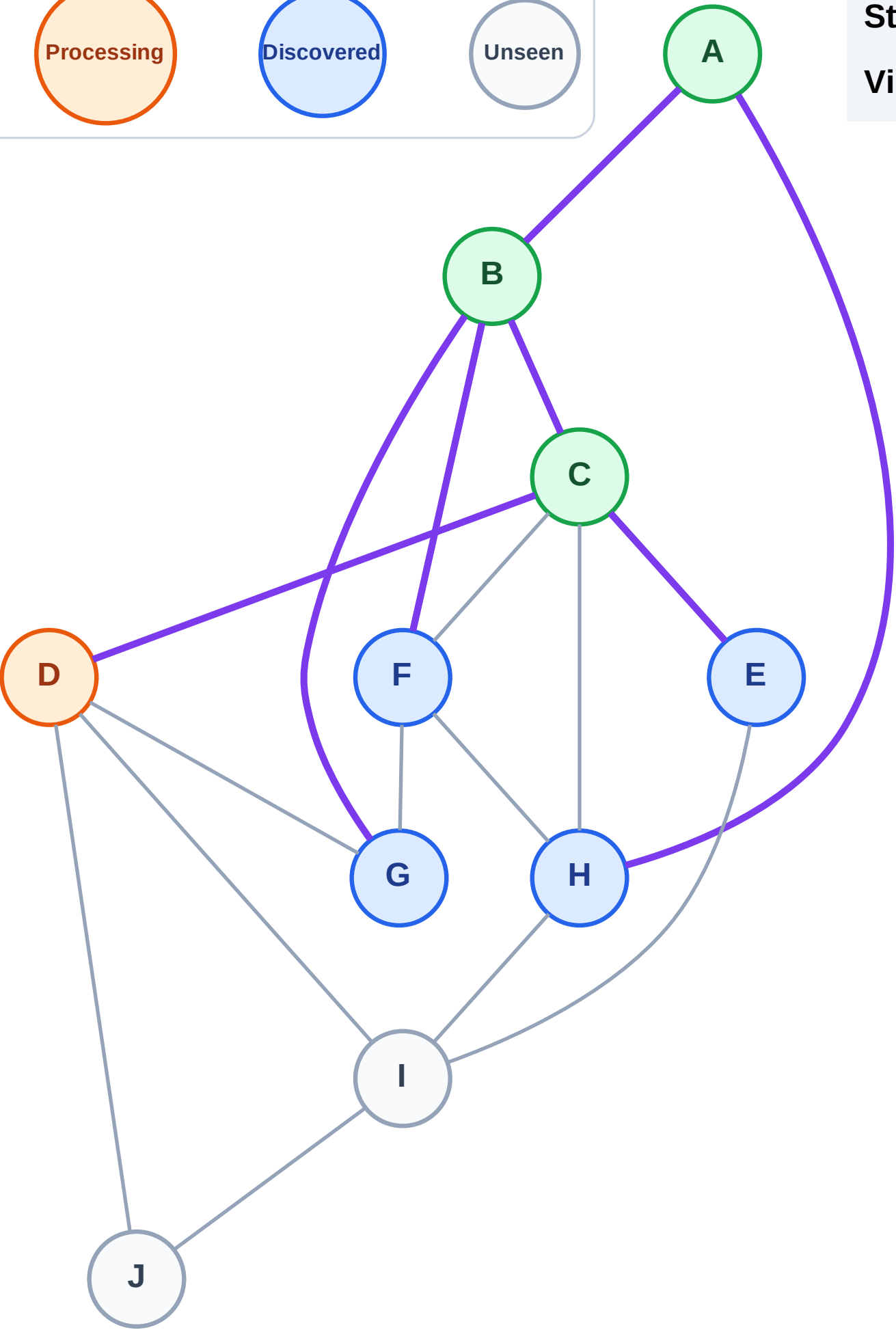
Complete

Processing

Discovered

Unseen

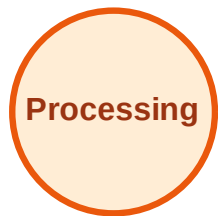
Stack (top at right): H | G | F | E
Visited: A, B, C



DFS Traversal

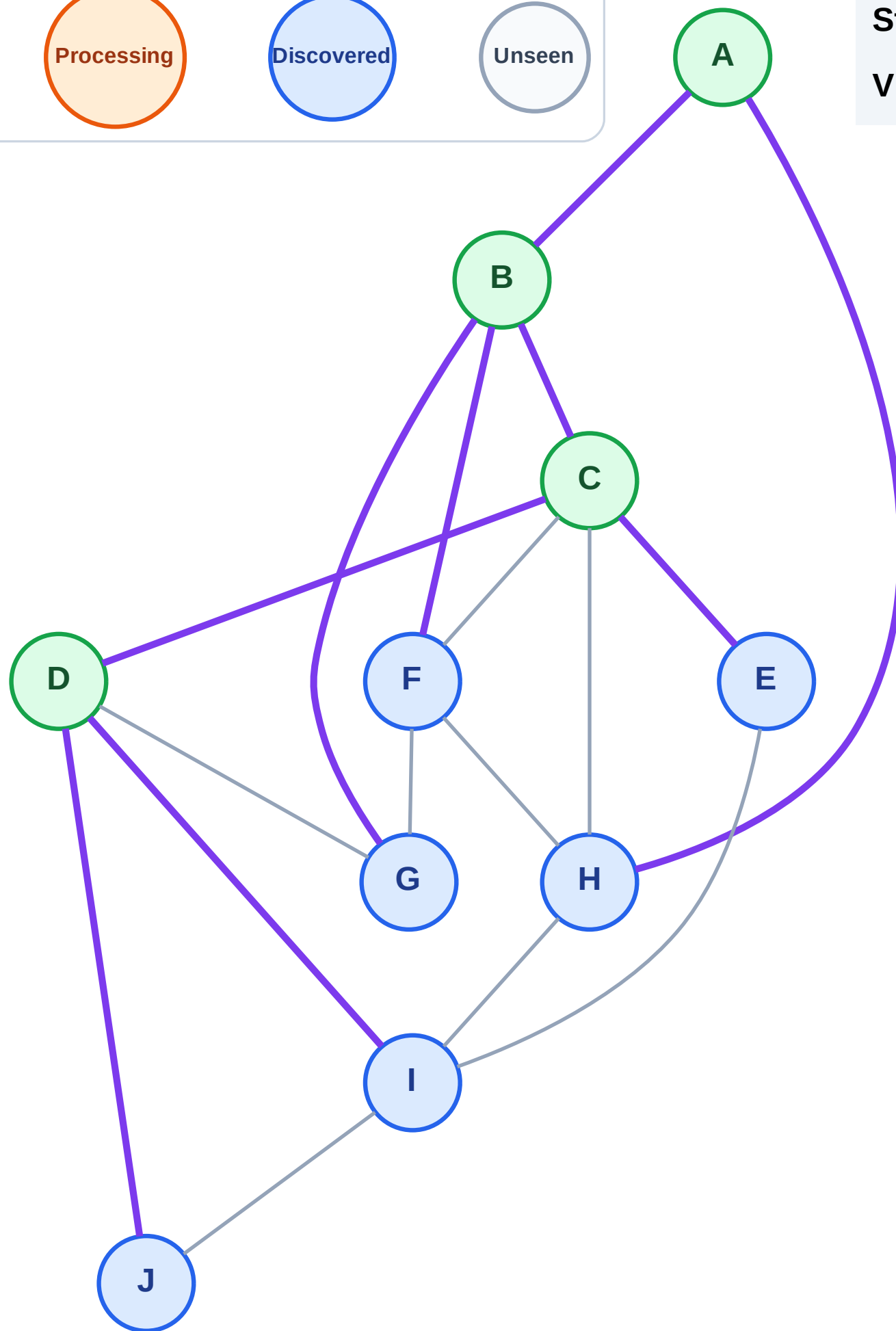
Step 9: Discover J, I from D

Legend



Stack (top at right): H | G | F | E | J | I

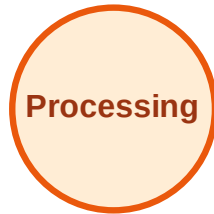
Visited: A, B, C, D



DFS Traversal

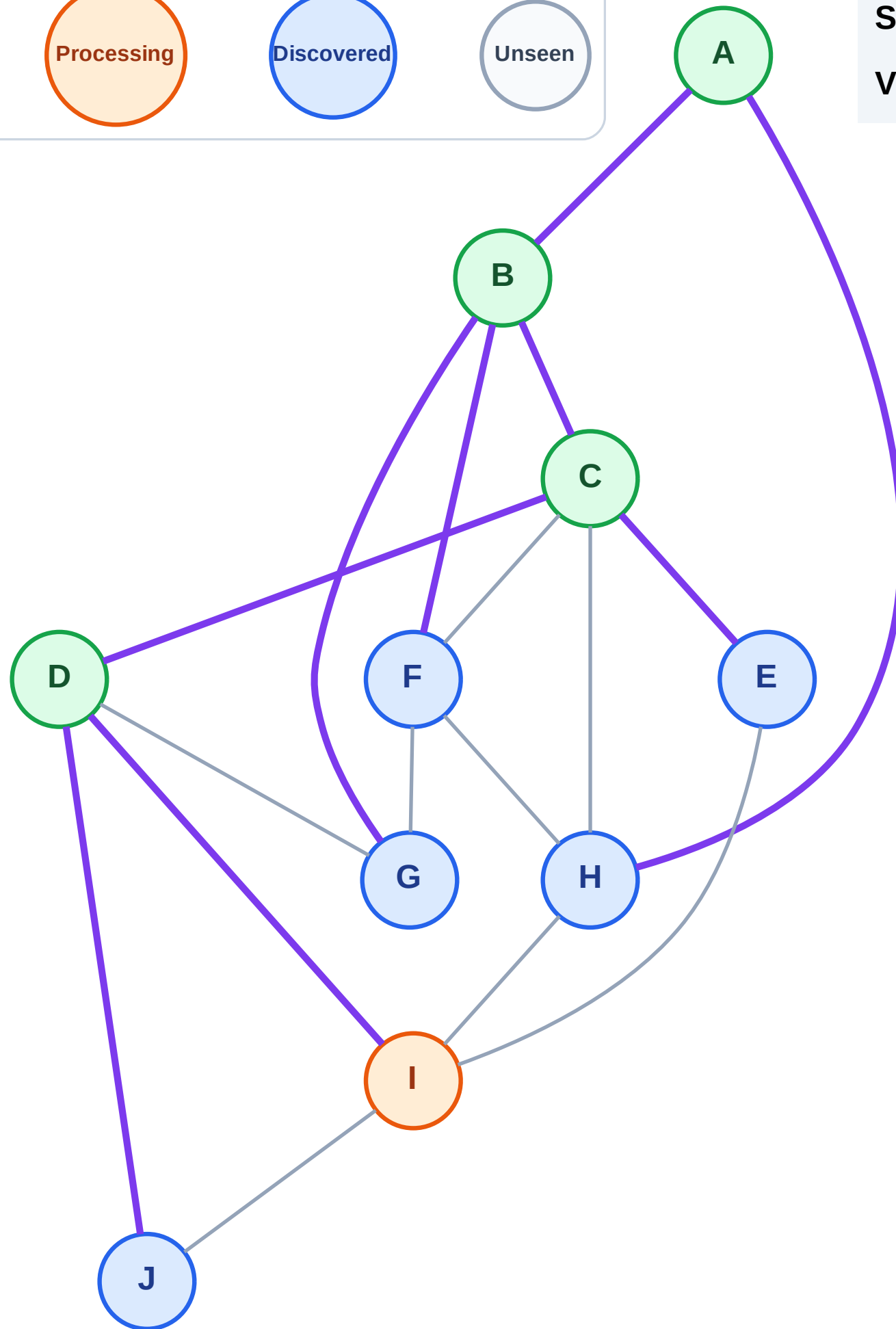
Step 10: Process node I

Legend



Stack (top at right): H | G | F | E | J

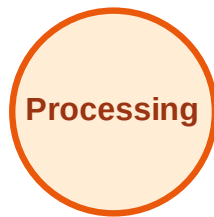
Visited: A, B, C, D



DFS Traversal

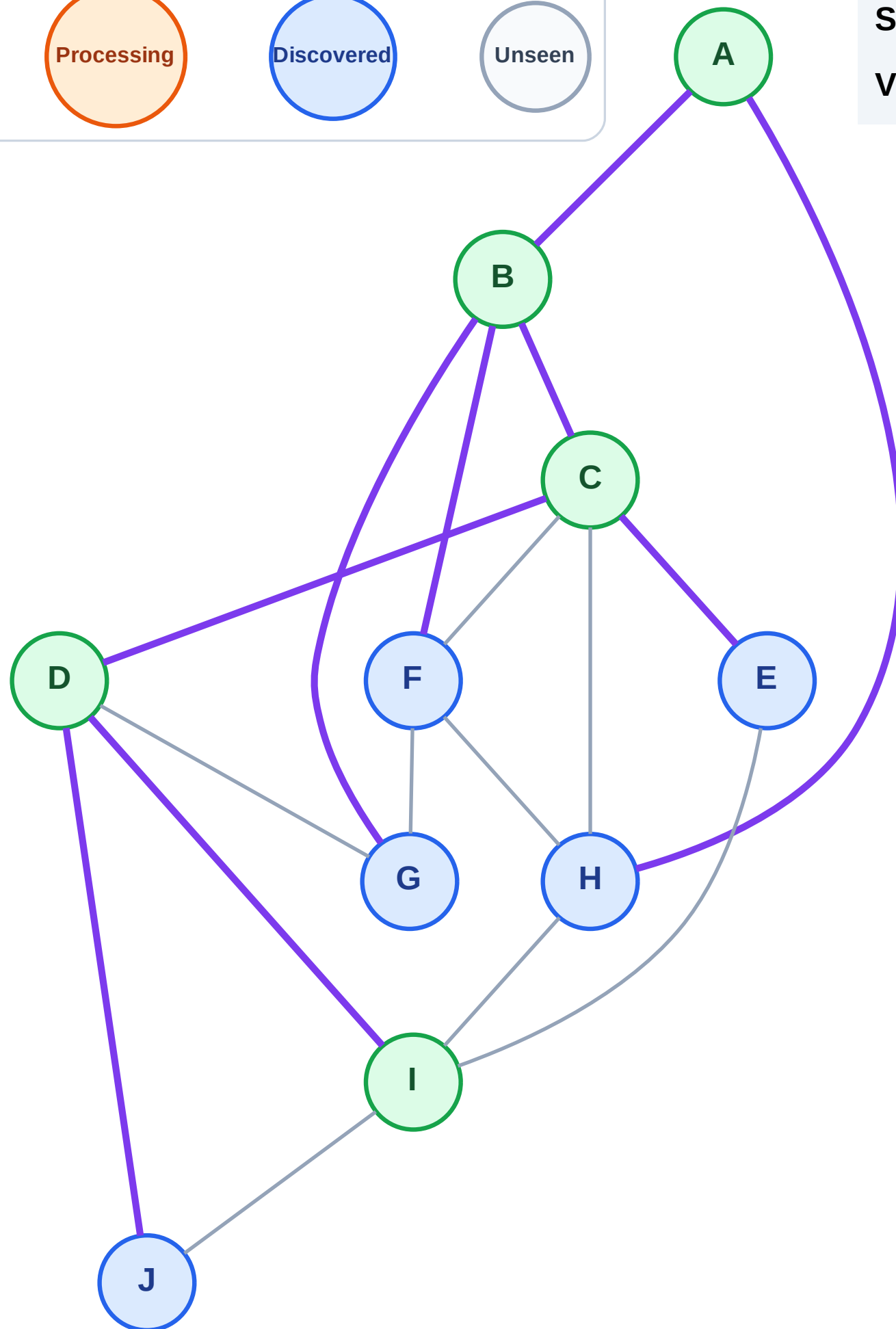
Step 11: No new neighbors from I

Legend



Stack (top at right): H | G | F | E | J

Visited: A, B, C, D, I



DFS Traversal

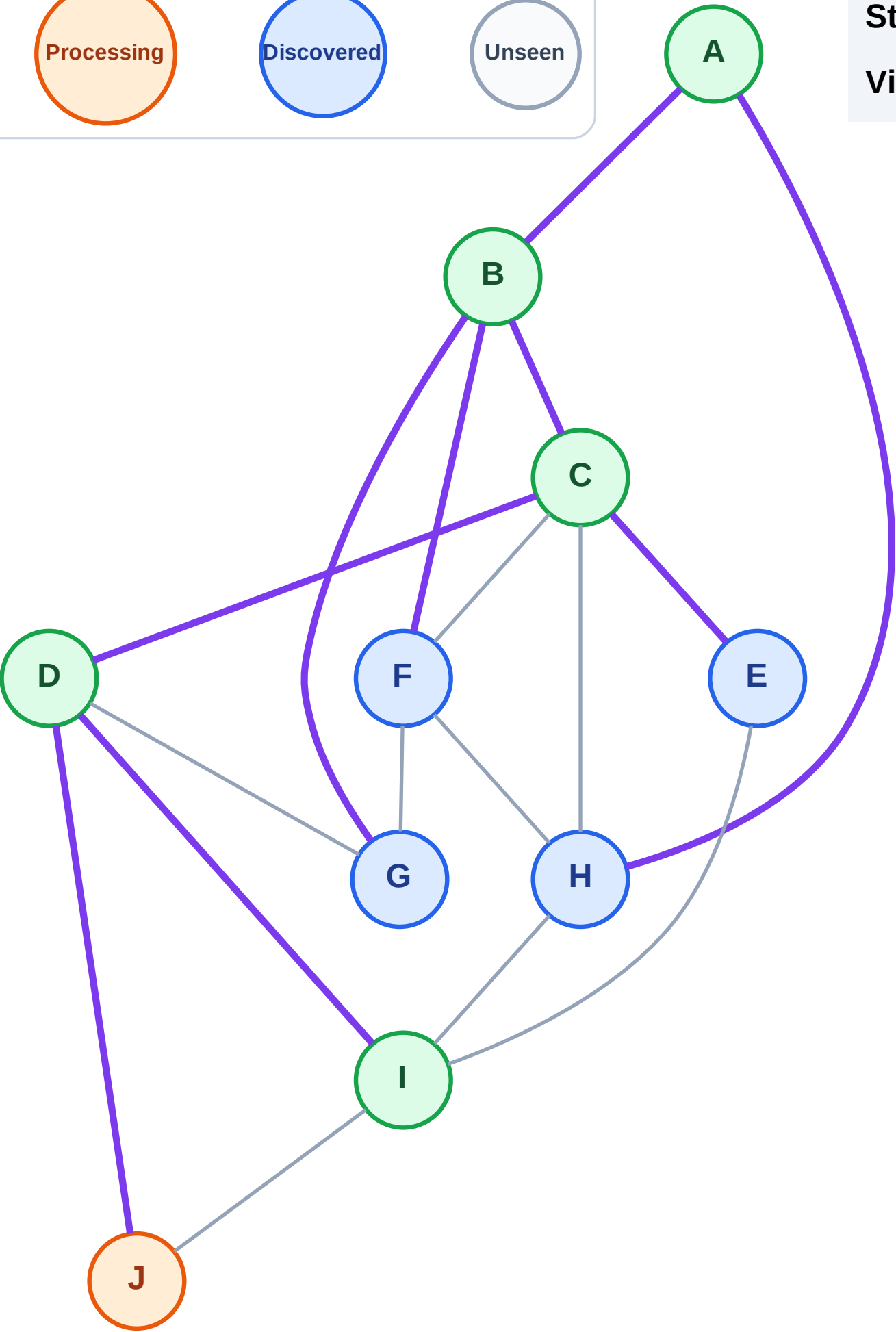
Step 12: Process node J

Legend



Stack (top at right): H | G | F | E

Visited: A, B, C, D, I



DFS Traversal

Step 13: No new neighbors from J

Legend

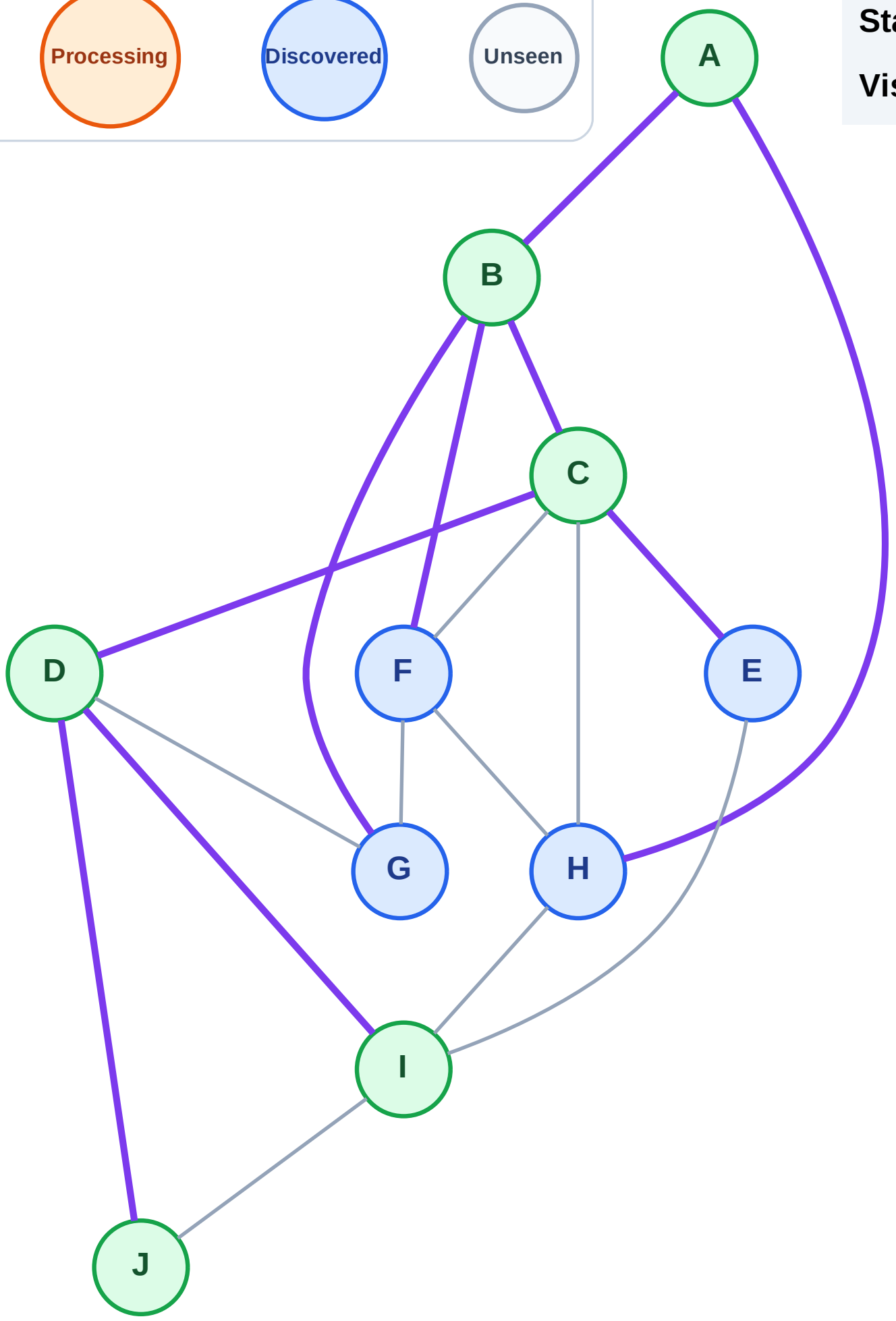
Complete

Processing

Discovered

Unseen

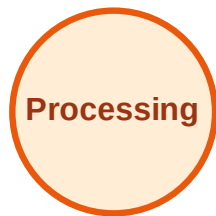
Stack (top at right): H | G | F | E
Visited: A, B, C, D, I, J



DFS Traversal

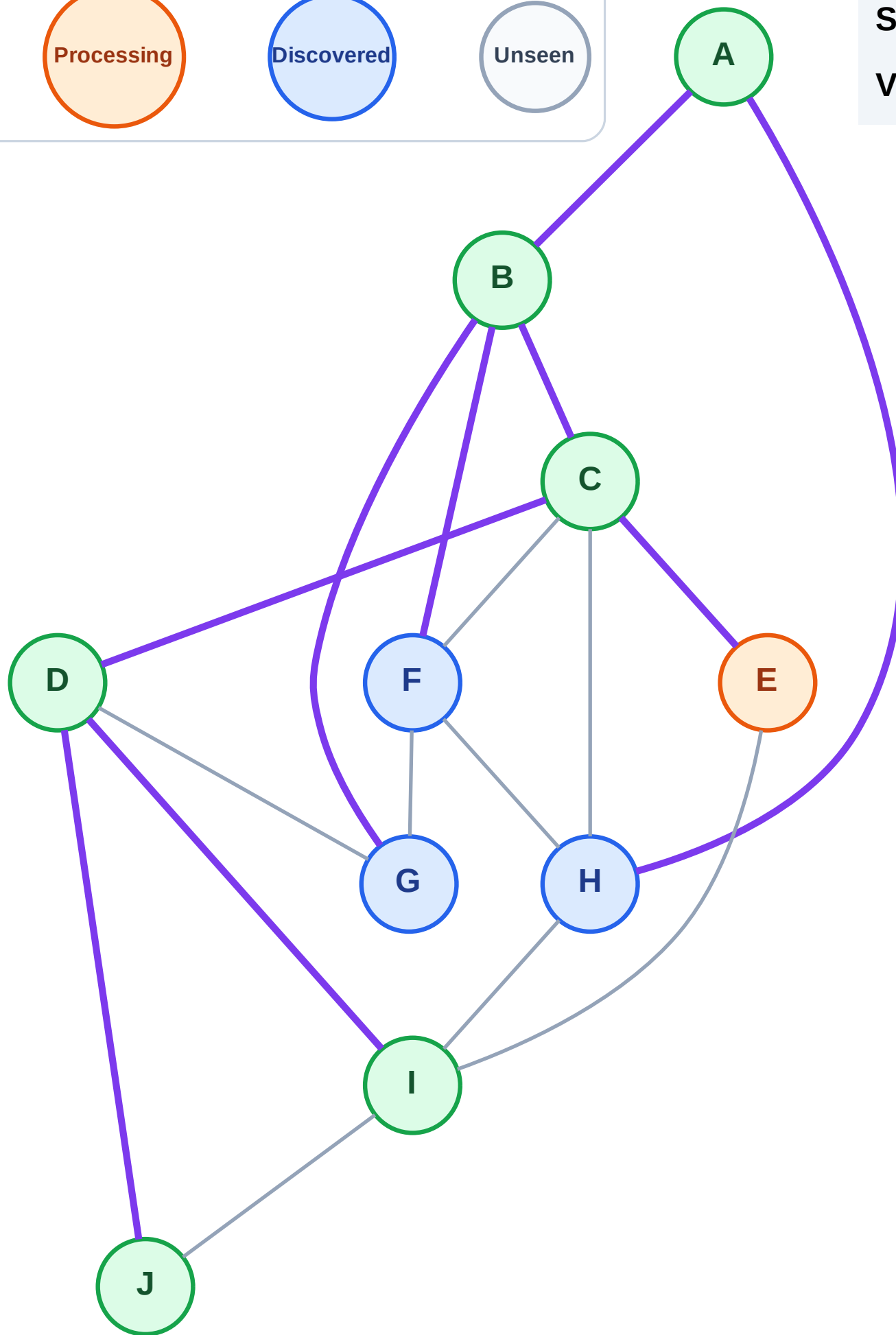
Step 14: Process node E

Legend



Stack (top at right): H | G | F

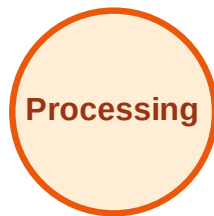
Visited: A, B, C, D, I, J



DFS Traversal

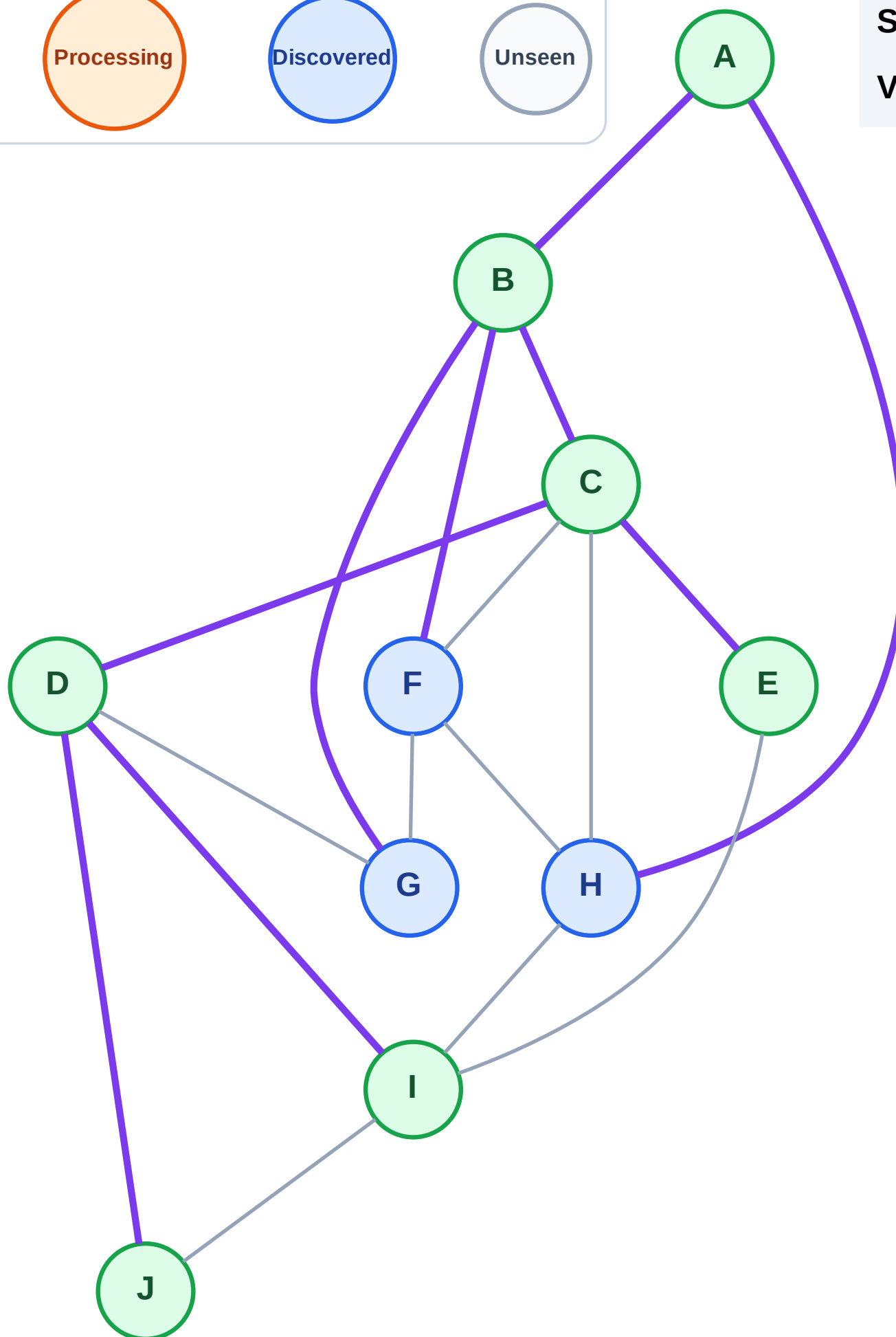
Step 15: No new neighbors from E

Legend



Stack (top at right): H | G | F

Visited: A, B, C, D, E, I, J



DFS Traversal

Step 16: Process node F

Legend

Complete

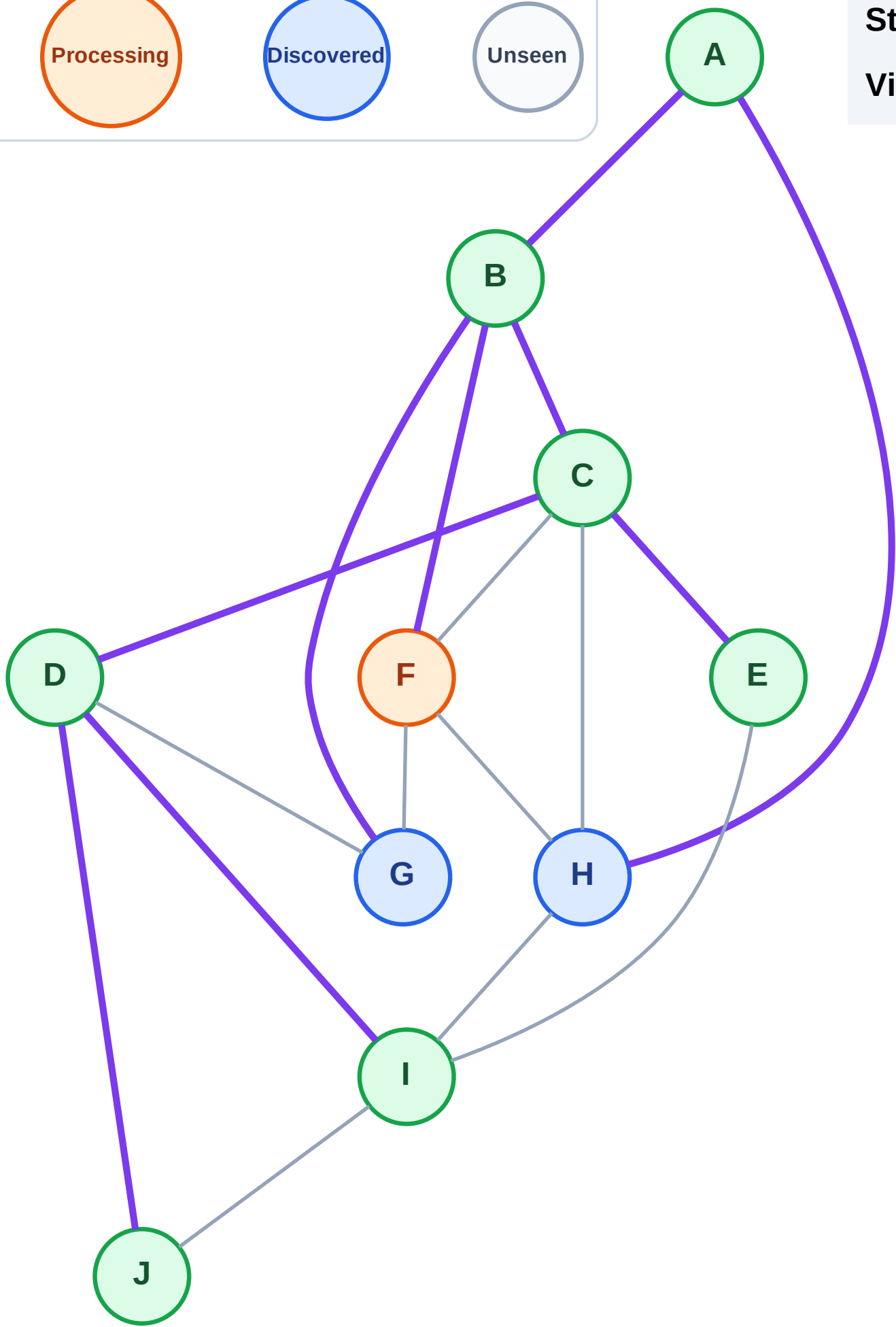
Processing

Discovered

Unseen

Stack (top at right): H | G

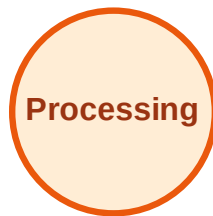
Visited: A, B, C, D, E, I, J



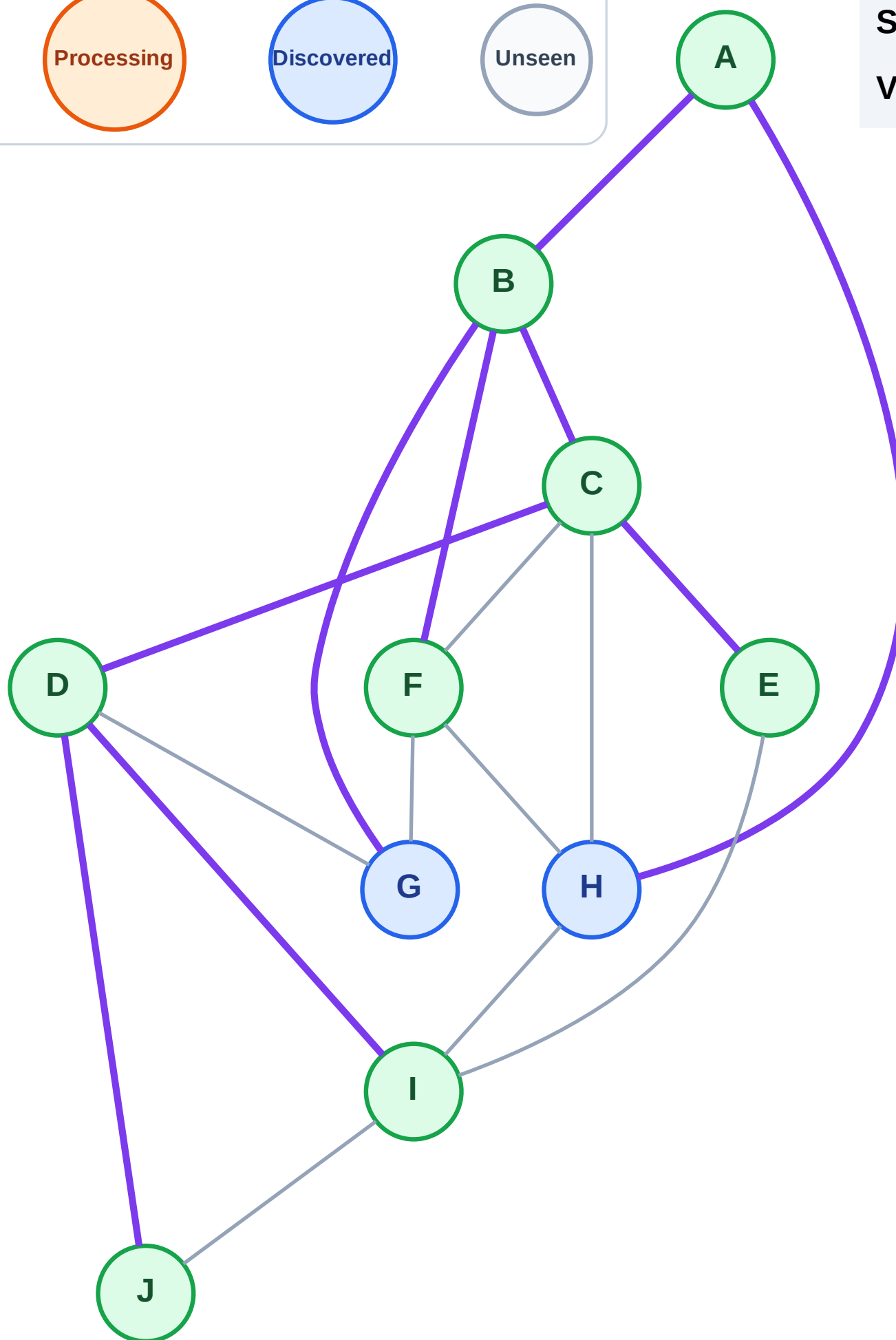
DFS Traversal

Step 17: No new neighbors from F

Legend



Stack (top at right): H | G
Visited: A, B, C, D, E, F, I, J



DFS Traversal

Step 18: Process node G

Legend

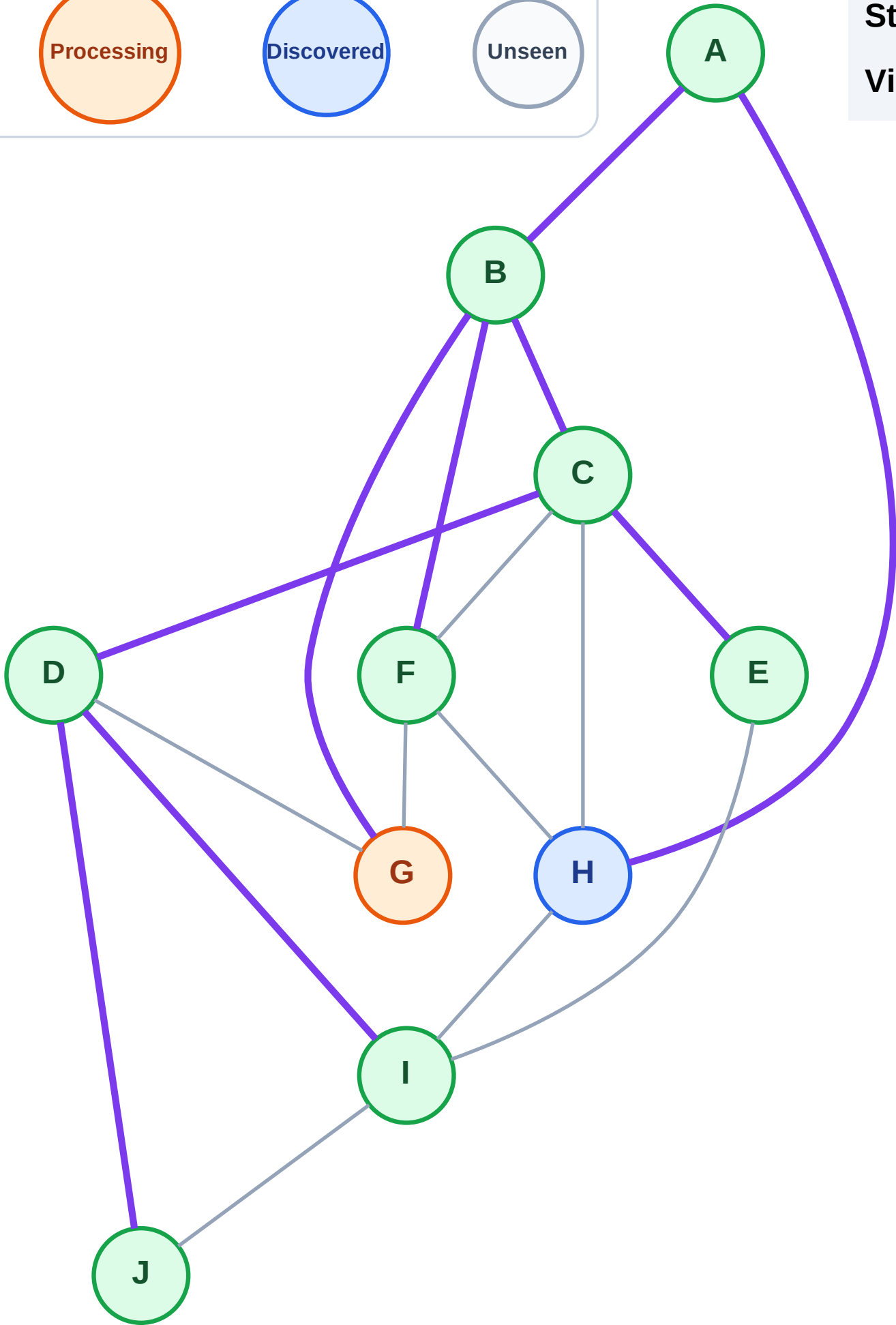
Complete

Processing

Discovered

Unseen

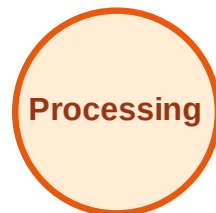
Stack (top at right): H
Visited: A, B, C, D, E, F, I, J



DFS Traversal

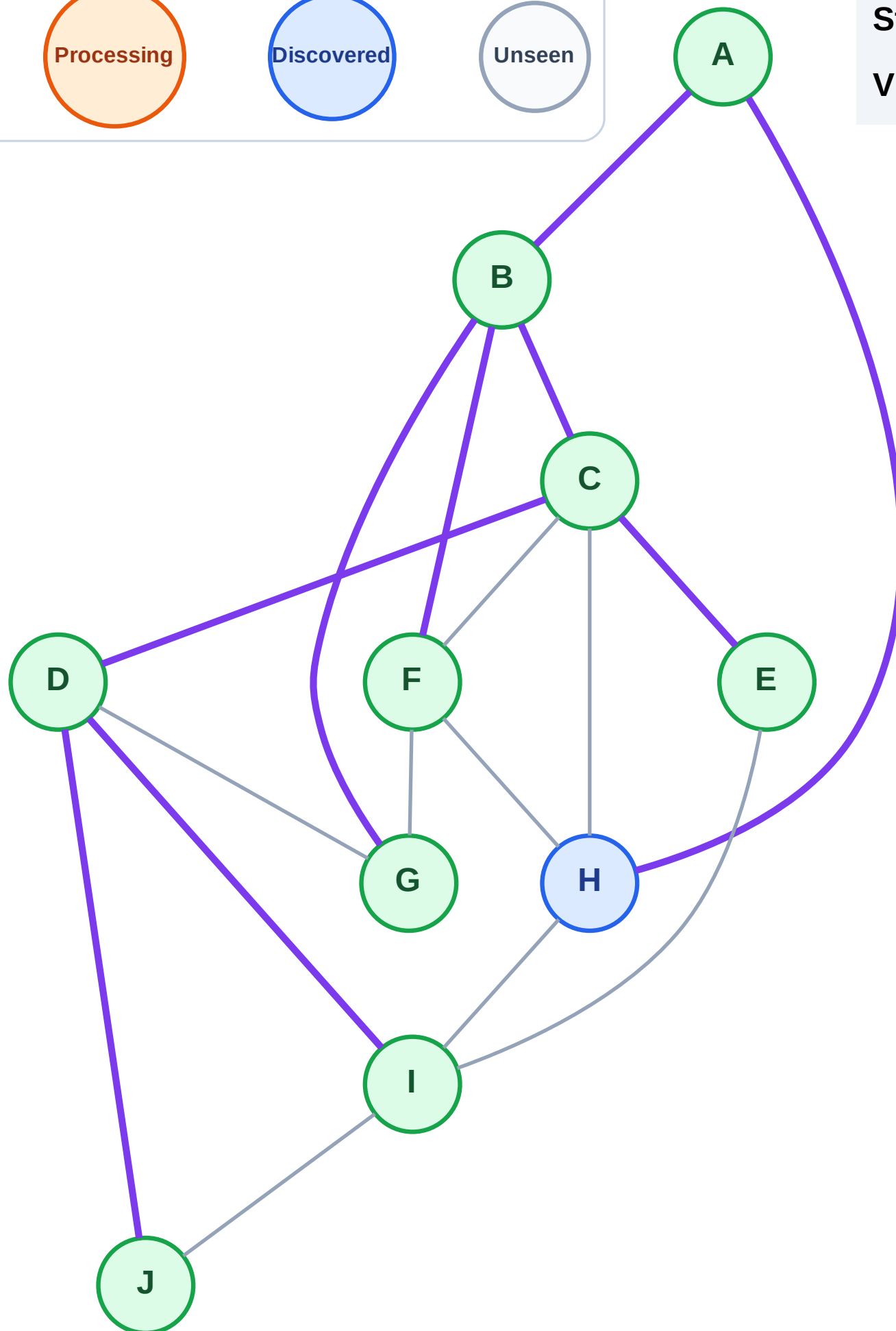
Step 19: No new neighbors from G

Legend



Stack (top at right): H

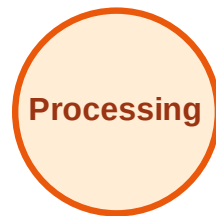
Visited: A, B, C, D, E, F, G, I, J



DFS Traversal

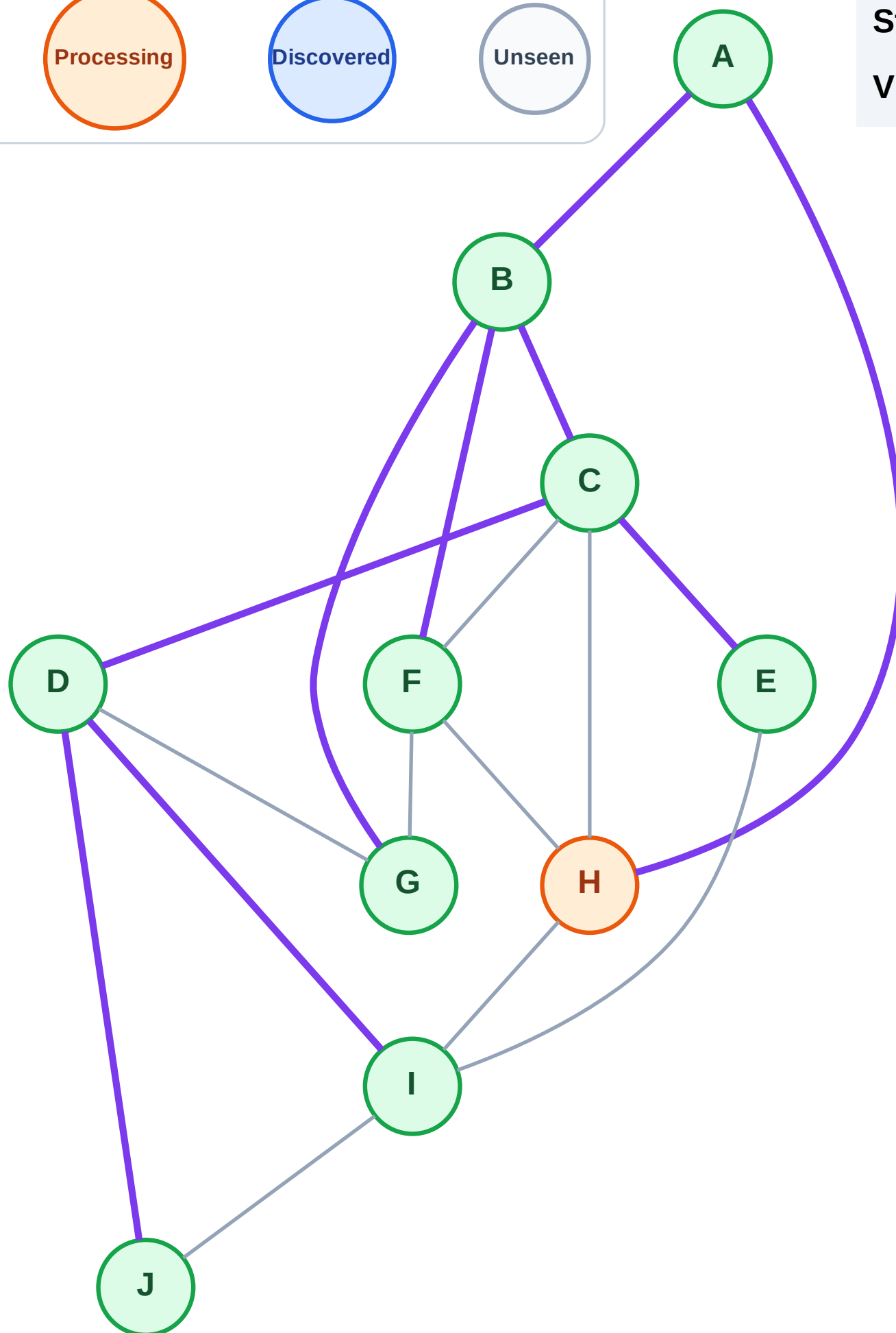
Step 20: Process node H

Legend



Stack (top at right): (empty)

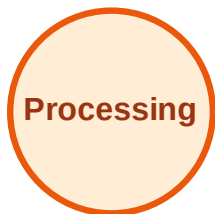
Visited: A, B, C, D, E, F, G, I, J



DFS Traversal

Step 21: No new neighbors from H

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

